

The 18GB Envelope and the Opus-4.7 Gap: An Honest Negative Result on Laptop-Scale Frontier-Equivalent Modeling

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Abstract—Program 1 of the AGI Lab asked an aspirational question: *can a model that is functionally equivalent to Claude Opus 4.7 fit on an 18GB M3 Pro laptop under current compression techniques and without cloud compute?* We operationalized this as a five-benchmark capability-breadth floor set (MMLU, HumanEval, GSM8K, GPQA Diamond, TruthfulQA) with a pre-declared falsifiability gate: does there exist *any* publicly-released $\leq 100\text{B}$ -total-parameter open-weight model, evaluated at FP16 or equivalent, that clears all five floors within a three-band proxy-aware policy? Version 1 of the gate (ARC-AGI-v2 primary) failed at orders-of-magnitude scale — the best eligible public model scores in the 1.3–4% range against Opus-4.7's 75.8%. We then ran a pre-committed one-shot benchmark swap (ARC-AGI-v2 $\rightarrow$ GPQA Diamond as primary, with ARC-AGI-v2 retained as a hard secondary floor at $\geq 5\%$) and re-ran the gate. The amended gate also failed: three-of-five primary hard-FAILs including a 43.7pp gap on GPQA Diamond (Llama 3.3 70B 50.5 vs Opus-4.7 94.2), robust to band widening up to $\pm 20\text{pp}$. A meta-pre-commitment bound us against a third relaxation; we honored it and closed Program 1 on this envelope paper. The paper makes three contributions: (1) a three-band proxy-aware falsifiability methodology suitable for capability-envelope programs where proxies are unavoidable; (2) a provenance-audited benchmark floor table for Opus-4.7 with explicit disclosure of generational trajectory; and (3) a worked example of pre-commitment discipline honored under adversarial pressure from an internal red team and from the program's own economic incentive to relax and proceed. The honest answer to the operational question under the current $\leq 100\text{B}$ public open-weights frontier and the 18GB unified-memory envelope is: *no instantiation-free path exists under current techniques.* Per lab value §9 (the mission is aspirational on purpose), this answer is the intended deliverable of an aspirational program, not a failure of apparatus.

Index Terms—feasibility envelope, falsifiability gate, negative-result methodology, proxy-aware benchmarking, MoE eligibility, compression frontier, pre-commitment, aspirational missions

A. What this paper is NOT (front-matter)

Before the body, three scoping disclaimers — carried forward from the P3 outline (alt_d_paper_outline.md) verbatim and preserved here because reviewer confusion on these three points is predictable and costly.

(a) This paper is not a Program-1-lite demo result. No model was trained on the 18GB laptop. No precision sweep, MoE trainability sweep, or scaled run was executed. The Program halted at Phase 2 (post-gate), consistent with the Alt-D branch of the program's scoping docu-

ment (scoping_1h.md). The contribution is the envelope characterization + methodology + provenance audit, not a from-laptop positive result.

(b) This paper is not a proposal for further SQ1 amendment. A meta-pre-commitment in sql_amendment_addendum.md §4 item 9 (PI-approved via the D-109 appendix to the amendment package) binds the program against further floor relaxation or eligibility-criterion amendment. Any reviewer who wants to see relaxed bands is directed to Appendix B and the companion artifacts; the paper declines to soften. The closure of Program 1 is the correct outcome; the correct follow-up to a reader who disagrees with the bands is a new program-open proposal, not an edit to this one.

(c) This paper is not a survey of $>100\text{B}$ models. The locked eligibility criterion is $\leq 100\text{B}$ total parameters, FP16 or equivalent, publicly-released open weights. Proprietary or closed or $>100\text{B}$ -total models appear only in the role of the Opus-4.7 reference floor itself (the target), not as candidates. Appendix D argues for the total-params reading over the active-params reading; the short version is that storage is the binding 18GB constraint, and a 671B-total MoE with 37B active does not fit at any precision regardless of routing sparsity.

B. §1 — Introduction

The AGI Lab opened Program 1 with an explicit aspirational question (programs/program_1_opus47_on_18gb/question.md:11):

“Can we fit an Opus-4.7-equivalent model on an 18GB M3 Pro laptop? What is the feasibility envelope given current techniques + 18GB unified memory + no cloud compute?”

The lab's values file (data/values.md §9) names this kind of question “aspirational on purpose.” The stated rationale is that “the mission exists to force the lab to develop capabilities it wouldn't otherwise build.” The program-open memo (program_open_memo.md:18-20) explicitly accepts that the outcome may be either (a) a rigorous envelope paper with gap-closing research directions or (b) a negative result with specific conditions

under which the mission is not currently achievable. Both are valuable; a negative result with clear conditions saves years of later work. Program 1 produced outcome (b). This paper is that outcome.

Phase 1 of the program ran four agents (chief_scientist, measurement_theorist, engineering_lead, evaluator) through a scoping + tractability + prior-work cycle, closed with a unanimous Principal-Investigator/Director lock (unanimous_lock_1i.md), and produced the operational question (question.md:15-17):

“On an 18GB M3 Pro laptop with no cloud compute, using current techniques (quantization, distillation, sparsity, gradient checkpointing), what is the maximum fraction of Opus-4.7’s published benchmark floors on {MMLU, HumanEval, GSM8K, ARC-AGI, TruthfulQA} achievable by (a) an inference-only deployment of a compressed existing SOTA model, and (b) a from-laptop trained/distilled model?”

Embedded in Phase 1’s sub-question decomposition (subquestions_1g.md SQ1) was a falsifiability clause requiring that before any engineering work, the benchmark set be tested for instantiation-freeness: “If no publicly-available $\leq 100\text{B}$ -parameter FP16 model clears all five locked floors, the benchmark set is ill-posed and must be narrowed/relaxed or the program closes on a negative-result paper.” This clause implements the lab’s value §1 (evidence outranks authority) and §8 (pre-commitment discipline, per the lab’s director prompt) before engineering work begins. The decision to build the gate into SQ1 itself, rather than discover unfeasibility downstream at P7 or P9, is the single most consequential methodological choice the program made, and most of the rest of this paper is the story of how that choice played out.

The findings of this paper can be stated in three sentences. First, Phase 2 (SQ1 Benchmark-Floor Pinning) ran the gate against the originally-locked five-benchmark set and FAILED by orders of magnitude on ARC-AGI-v2 (Opus-4.7 75.8% vs the best $\leq 100\text{B}$ public model at $\sim 1.3\%$). Second, after a pre-committed one-shot amendment (ARC-AGI-v2 $\rightarrow$ GPQA Diamond as primary, with ARC-AGI-v2 promoted to a hard secondary floor at $\geq 5\%$) and a pre-declared three-band proxy policy, the re-run gate also FAILED, with the GPQA Diamond gap alone at 43.7pp and a secondary-floor FAIL on ARC-AGI-v2 — independently. Third, a meta-pre-commitment against any further relaxation was honored; Program 1 closed on this paper rather than on a pivot to laptop-scale distillation, a softened floor set, or an active-params MoE eligibility reading that would have reintroduced the gap at a different dimension.

What a negative result means here. The finding is not “our apparatus broke” nor “our trainer was too small” nor “we ran out of sessions.” The finding is about a specific public frontier: the $\leq 100\text{B}$ FP16 open-weight state of the public frontier as of 2026-04-17 does not support

the target Opus-4.7 floor vector, and the 18GB unified-memory envelope compounds this constraint rather than relieving it. A future $\leq 100\text{B}$ FP16 public model that cleared the amended primary set would reset the question (sq1_benchmark_floors_v2.md §6.5). This paper characterizes the gap precisely enough that such a future assessment is straightforward to run. Until such a model exists, Program 1’s answer to the operational question is “zero instantiations exist.”

The rest of the paper proceeds as follows. §2 describes the hardware envelope, the benchmark set, and the eligibility + three-band policy, and in §2.5 situates the paper against prior negative-result / feasibility-envelope / pre-registration methodology literature. §3 reports the v1 falsifiability failure on ARC-AGI-v2. §4 reports the v2 failure on the amended set. §5 surveys the published 18GB-envelope implications for compression, MoE, and sparsity — without running a from-laptop sweep. §6 frames the result in the lab’s aspirational-mission terms (values §9). §7 documents the methodological contributions (three-band policy, protocol-asymmetry handling, MoE eligibility argument, pre-commitment discipline) and points to the appendices that carry full technical detail. §8 closes the program with forward pointers. Appendices A–E carry the floor provenance audit, the three-band policy specification, the protocol-asymmetry corrections, the MoE total-vs-active argument, and the publication-decision stub.

C. §2 — Background and Setup

1) 2.1 Hardware envelope: The 18GB M3 Pro laptop is the lab’s single piece of compute (CLAUDE.md / project instructions). It is a unified-memory architecture: CPU + 18 GPU cores + Neural Engine + system share one physical 18GB pool. The M3 Pro GPU runs at SIMD width 32 and is the only GPU surface the lab’s Metal shader pipeline targets. The hard working-set ceiling for inference or training is approximately 16GB after OS reservation + KV cache + activations + optimizer state + I/O buffers (question.md §Resource budget). Disk is 119GB local plus $\sim 1.3\text{TB}$ iCloud (chunked streaming per data/procedures.md:140). There is no cloud compute, no external APIs, no pay-per-call inference endpoints — this is a stated constraint, not a practical inconvenience.

The consequence for this paper is mechanical. A model that must be loaded to RAM in order to run inference is bounded by that $\sim 16\text{GB}$ working-set at its chosen numeric precision. At Q4 (4-bit weights), that ceiling admits roughly 36B dense parameters before activation and KV budget; at INT8 it admits $\sim 18\text{B}$; at FP16 it admits $\sim 9\text{B}$. These ceilings are well below the $\leq 100\text{B}$ eligibility bar for the falsifiability gate — in other words, even if the public frontier did produce a $\leq 100\text{B}$ model that cleared the floors, the envelope question would then reduce to whether a further 2–3 $\times$ compression step could preserve those floor scores inside 16GB. §5 returns to this compounding.

2) 2.2 “Equivalent” defined via benchmark fraction, not parameter count: Parameter-count equivalence to Opus-4.7 is explicitly rejected. Opus-4.7 is widely estimated (through indirect disclosures and published compute budgets — see prior_work_1e.md:23 and references in §2 of that document) to be a sparse MoE model with trillions of total parameters and tens-to-hundreds of billions of active parameters per token. A 36B dense Q4 model is smaller than Opus-4.7 by roughly three orders of magnitude in total parameter count. The program-open memo and the locked question.md therefore define equivalence functionally, via the fraction of Opus-4.7’s published benchmark floors a candidate configuration can clear on a standardized capability-breadth set.

This decision has a methodological cost that the rest of the paper pays down. Once “equivalent” is a floor-fraction, two things become necessary: (a) the floor values must actually be Opus-4.7’s scores (proxy inheritance from sibling models is accepted only with explicit marking), and (b) a scoring policy must exist for values that are close but not at the floor. The provenance audit in Appendix A addresses (a); the three-band proxy-aware policy in Appendix B addresses (b).

3) 2.3 Locked benchmark set (post-amendment): The primary five-benchmark set as it stands post-amendment (sql_benchmark_floors_v2.md §2):

- MMLU (5-shot) — Opus-4.7 floor 91.0% (proxy-inherited, ± 1.5 pp band).
- HumanEval (pass@1) — Opus-4.7 floor 96.8% (third-party reported, ± 1.0 pp band).
- GSM8K (maj@1 or 5-shot CoT) — Opus-4.7 floor 96.0% (proxy-inherited from saturation, ± 2.0 pp band).
- GPQA Diamond — Opus-4.7 floor 94.2% (direct Anthropic publication + multi-source corroboration, ± 3.0 pp band).
- TruthfulQA (MC2) — Opus-4.7 floor 65.0% (proxy-inherited, ± 3.0 pp band).

And the hard secondary floor:

- ARC-AGI-v2 — $\geq 5\%$ absolute (threshold, not proxy; ± 0.5 pp band). The threshold itself is below Opus-4.7’s reported 75.8%; a 5% floor tests whether any ≤ 100 B public model has non-trivial reasoning signal on v2, rather than whether any achieves Opus-4.7-equivalence on it. This secondary floor is the “Attack 3 resolution” from the red-team review, promoting ARC-AGI-v2 from cosmetic diagnostic to a binding constraint (sql_amendment_addendum.md §2 Attack 3 disposition; §4 item 2).

The post-amendment set differs from the originally-locked set in exactly one primary swap — ARC-AGI-v2 $\rightarrow$ GPQA Diamond — plus the promotion of ARC-AGI-v2 to secondary. No band was widened post-hoc. The amendment narrative and its PI co-sign are recorded at §4 below.

4) 2.4 Eligibility criterion (locked, total-parameters reading): A candidate “public ≤ 100 B FP16 open-weight model” is, for the purposes of the gate:

1. Parameter count: ≤ 100 B total parameters. This includes dormant MoE experts, unused adapters, and any other weights required to run inference. The argument for total-params over active-params is in Appendix D; the short version is that storage is the binding 18GB constraint, and a 671B-total MoE (e.g., DeepSeek V3.2 at 37B active, 671B total) occupies ~ 336 GB at Q4 — $\sim 20\times$ larger than the 18GB physical pool, even ignoring activation and KV overhead.
2. Precision: FP16 or equivalent published number. Quantized-to-Q4 scores are accepted as informational but not as the candidate’s headline floor score, because quantization-induced score loss is itself an object of study in §5 and would be a confounding variable in the gate.
3. Release: publicly-released weights, downloadable, runnable without a gated vendor API. A closed model (e.g., Opus-4.7 itself, o3, Gemini 2.5 Pro) cannot be a candidate — it can only be the target floor. An announced-but-not-released model is not a candidate.

Under this reading, the eligible list as of 2026-04-17 includes Llama 3.1/3.3 70B, Qwen2.5-72B and its coder/math variants, Qwen3 32B dense and QwQ-32B-preview, smaller Mistral/Llama/DeepSeek dense variants ≤ 100 B, Mistral-Large-123B fails the threshold (> 100 B total). Under the permissive active-params reading (Appendix D steelman), the list expands to include DeepSeek V3/V3.2 (671B total / 37B active), DeepSeek R1 (671B total / 37B active), Qwen3.5-235B-A22B MoE (235B total / 22B active), and some Mixtral variants. Our gate runs under the total-params reading; §6.2 of the floor file also reports results under the active-params reading for transparency; both readings fail, but total-params fails decisively.

5) 2.5 Related Work (new in v2; positioning against methodology literature): Program 1’s methodological contribution is a pre-registered falsifiability gate attached to an aspirational capability-envelope program, combined with a three-band proxy-aware scoring policy and a meta-pre-commitment that forecloses goalpost-movement after an authorized one-shot relaxation. This positioning intersects four strands of prior literature.

Negative-result methodology in ML. The “I Can’t Believe It’s Not Better” (ICBINB) workshop series at NeurIPS has curated honest-negative-result submissions since 2020, establishing a normative frame in which a well-characterized non-result is publishable and cumulative. The workshop’s explicit criteria include (i) a clearly-stated hypothesis, (ii) a falsification protocol specified before running the experiment, and (iii) a forward-pointing discussion of what conditions would change the

verdict. This paper satisfies all three: the hypothesis is the operational question (question.md), the protocol is the SQ1 falsifiability clause with the Reading B gate criterion, and the forward-pointing conditions are enumerated in §6.4 + §8.4.

Pre-registration in ML. Pre-registration of experimental protocols — standard in psychology and drug trials since the early 2010s — has been proposed for ML evaluation by van Miltenburg et al. and others. Implementing pre-registration in ML evaluation requires that (i) the evaluation metrics, (ii) the model cohort, and (iii) the pass/fail criteria be fixed before any candidate model is scored. Program 1 implements this via the SQ1 lock, the $\leq 100\text{B}$ eligibility criterion, and the three-band policy all being sealed before Phase 2 scoring.

Feasibility-envelope methodology. Resource-constrained deployability studies (edge-device ML, on-device inference, mobile deployment) have converged on framings that separate (a) model-level capability claims, (b) hardware-envelope fitness claims, and (c) compression-path feasibility claims. Prior work in this tradition typically runs one or more partial-positive experiments; this paper stops at (a) because (a) fails the falsifiability gate, which is the single most cost-efficient place to stop. The methodology contribution is specifically that the envelope can be characterized negatively without running (b) or (c), by characterizing the frontier starting-point gap against the target.

Reproducibility crisis and benchmark proxy-inheritance. The reproducibility discussion in ML (Pineau et al., NeurIPS 2019 reproducibility checklist; Gundersen et al.; Hutson 2018) identified benchmark-score volatility across reproductions as a primary confounder. Proxy inheritance — using sibling-model scores as a floor for a non-scored target — magnifies this volatility. The three-band policy (Appendix B) is a concrete proposal for handling proxy-inheritance uncertainty without enabling post-hoc band widening, via per-benchmark pre-declaration of band widths motivated by noise-floor arguments rather than by desired-outcome arguments.

No published paper known to us combines (a) falsifiability-gate pre-registration with (b) three-band proxy-aware scoring with (c) a meta-pre-commitment against serial relaxation. This combination is the methodological claim of §7; the prior-art situates the individual components but not the composite.

D. §3 — Falsifiability Failure v1: ARC-AGI-v2

1) 3.1 The original v1 floor: The five-benchmark floor set as originally locked in Phase 1 (sq1_benchmark_floors.md §2 — preserved under §4 archive-never-delete despite subsequent amendment) included ARC-AGI-v2 as its primary reasoning-ceiling signal. ARC-AGI-v2 is the 2024 refresh of the ARC-AGI challenge by the ARC Prize foundation; it is structured as semi-private evaluation where public models are evaluated against private test

items, with leaderboard entries vetted by the organizers. Scores are reported as percentages of correct solutions.

Opus-4.7 on ARC-AGI-v2: 75.8% at max-effort protocol, directly published by Anthropic and corroborated on the ARC Prize leaderboard under Anthropic’s submissions for the Claude 4-family. This is the highest-confidence numeric floor in the original v1 set — single primary source, protocol disclosed, multi-source corroboration. sq1_benchmark_floors.md §2 ARC-AGI-v2 row records the confidence as HIGH.

2) 3.2 Best $\leq 100\text{B}$ FP16 public candidate on ARC-AGI-v2: We surveyed the public ARC Prize leaderboard and the published evaluation harness outputs of each eligible $\leq 100\text{B}$ model as of the Phase 2 gate date (2026-04-17). The strongest public $\leq 100\text{B}$ reasoning-trained candidate on ARC-AGI-v2 is DeepSeek R1 — note that R1 is ineligible under the total-params reading (671B total), but even under the permissive active-params reading which would admit it, R1’s public ARC-AGI-v2 score is approximately 1.3% (ARC Prize leaderboard, R1 submission on v2 public-eval).

Moving to the total-params-eligible list: QwQ-32B-preview from Alibaba’s Qwen team is Alibaba’s reasoning-trained $\leq 100\text{B}$ dense model and has been evaluated on v2 with leaderboard-reported scores in the 1–4% range across evaluation protocols. No other eligible $\leq 100\text{B}$ model has a published ARC-AGI-v2 score that exceeds 4%. The collective upper bound of the eligible-candidate ARC-AGI-v2 score as of 2026-04-17 is therefore approximately 4%.

3) 3.3 Gap magnitude and decision: The absolute gap on ARC-AGI-v2 under the v1 set is $\sim 74\text{pp}$ (75.8% floor — 1.3% best-eligible-candidate-R1 under active-params reading, or 75.8% — $\sim 4\%$ QwQ under total-params reading). This is approximately two orders of magnitude in the failing direction on the fractional-correct scale. No defensible Reading-B band closes this gap; even a hypothetical Reading-C $\pm 20\text{pp}$ band (which we would not have adopted on §8 grounds, but consider here purely as a sensitivity check) leaves a gap of $\sim 54\text{pp}$. The failure is not a band-calibration issue. It is a statement about the reach of the public $\leq 100\text{B}$ open-weight frontier on ARC-AGI-v2 as of 2026-04-17.

Decision consequence. Phase 2 gate v1 FAILED. Per the SQ1 falsifiability clause (subquestions_1g.md SQ1), the locked set was declared ill-posed under the $\leq 100\text{B}$ FP16 public-model framing. This triggered the SQ1 amendment protocol — a pre-authorized path in the falsifiability clause itself, not a moving-goalposts move by ambush. The Phase 2 report (sq1_benchmark_floors.md §4 “Falsifiability Test”) explicitly applied the pre-declared threshold and reported the FAIL truthfully. An independent evaluator run (data/evaluator_report.md Critical Issue A) confirmed that pre-commitment was honored: measurement_theorist did not soften the ARC-AGI-v2 floor, did not narrow the candidate list, and did not adjust the band widths — it reported the hard orders-of-magnitude gap as-is. Phase 2

closed PASS_WITH_FLAGS, which in the lab’s grammar means “the phase did its job; the answer is that the locked set is ill-posed; Phase 3 is blocked pending amendment.”

4) 3.4 Framing: a claim about the frontier, not about the lab: The v1 finding is not about Program 1’s apparatus. No from-laptop run contributed to it. The finding is the statement: the public $\leq 100B$ FP16 open-weight frontier’s reasoning performance on ARC-AGI-v2 is approximately two orders of magnitude below Opus-4.7’s published performance. No 18GB laptop-side compression or distillation could close a two-orders-of-magnitude gap by construction under current techniques — compression monotonically loses quality, it does not add it. The v1 finding therefore establishes, before any compression work is done, that under the v1 benchmark set any from-laptop path is blocked because there is no $\leq 100B$ public model to start from that has the target property.

The v1 finding on its own would have been sufficient to close Program 1 on an envelope paper. We chose instead to run a one-shot amendment and a gate re-run, for three reasons: (1) the SQ1 falsifiability clause explicitly authorized a narrow relaxation in advance, so a single swap would not itself violate §8; (2) a reviewer could reasonably wonder whether the v1 finding was ARC-AGI-v2-specific (e.g., ARC Prize eval is semi-private; perhaps that evaluation protocol alone disadvantages $\leq 100B$ public models); (3) the amendment would clarify whether the v2 gate failed for benchmark-selection reasons or for frontier-gap reasons. §4 reports that the v2 gate also failed, at a different benchmark with fully public evaluation, which falsifies (2) and settles (3). The envelope finding therefore rests on two independent falsifications rather than one.

E. §4 — Falsifiability Failure v2: Amended Set (GPQA Diamond)

1) 4.1 The one-shot amendment: Director drafted an amendment proposal (sq1_amendment_proposal.md, Director-signed 2026-04-17). The proposal enumerated the three relaxation options surfaced by measurement_theorist at Phase 2 close — (1) relax ARC-AGI-v2 to a public-model-achievable level (e.g., 30%), (2) drop the $\leq 100B$ FP16 precondition entirely, or (3) swap ARC-AGI-v2 for a less-saturating reasoning benchmark with a tractable public-model-vs-Opus gap. Director rejected (1) on goalpost-movement grounds and (2) on “evades the check rather than resolving it” grounds, and recommended sub-option 3b: swap ARC-AGI-v2 for GPQA Diamond.

An internal red-team review (data/engineering/red_team_amendment_review_2026-04-17.md) then stress-tested the amendment. Two HIGH-severity findings resulted: Attack 3 argued that “retaining ARC-AGI-v2 as diagnostic” without any binding floor was cosmetic and should be upgraded to a hard secondary floor; Attack 4 argued that even an authorized one-shot relaxation fits the pattern §8 exists to prevent, and that a meta-pre-commitment should be added binding the program against

any further SQ1 amendments regardless of future outcome. Director accepted both in the amendment addendum (sq1_amendment_addendum.md §2 Attacks 3 and 4; §4 items 2 and 9). chief_scientist separately formalized the three-band proxy-aware policy (p1_4_proxy_acknowledgment.md §3) and that was also integrated into the addendum.

A self-critique by Director (sq1_amendment_gate_criterion_supplement.md) then surfaced an internal inconsistency: the parent proposal’s “Expected outcome: pass (within 10-20pp)” footnote was nowhere a formally-declared gate criterion. The supplement enumerated three readings of the gate criterion — Reading A (strict original, no bands), Reading B (three-band extension), Reading C (± 10 -20pp as implied by the footnote) — and recommended Reading B on consistency-and-pre-commitment grounds.

The amendment package (proposal + addendum + supplement + proxy policy) was submitted to PI review. The D-109 decision recorded a real PI-agent dispatch returning APPROVE under Reading B. The PI explicitly rejected Reading C on §8 grounds (adopting a permissive band silently at gate time is post-hoc criterion-shopping, regardless of signature) and rejected Reading A on consistency grounds (the three-band policy was already locked for SQ2-SQ5, so the gate re-run should use the same bands). The PI commended the Director self-critique as the correct pre-gate behavior for surfacing the ambiguity before PI signed. See data/agents/pi/episodic/2026-04-17_program_1_phase_1_and_2.md Part B for the full rationale.

2) 4.2 The pinned GPQA Diamond floor: GPQA Diamond is the hardest 198-question subset of GPQA (Rein et al., GPQA: A Graduate-Level Google-Proof Q&A Benchmark, arXiv:2311.12022, 2023), comprising multiple-choice physics/chemistry/biology questions written by domain PhDs and filtered against expert and non-expert solvers (Diamond = questions with $\geq 2/3$ expert agreement AND $\geq 2/3$ non-expert error). The standard evaluation protocol is 0-shot Chain-of-Thought with single-answer output, though vendors occasionally report maj@k (majority voting across k samples) and/or extended-thinking variants.

Opus-4.7 on GPQA Diamond: 94.2%. Source set:

- Anthropic launch/system-card materials for Claude Opus 4.7 (reported figure 94.2%).
- Vellum model-card aggregation (94.2% listed for Claude Opus 4.7 GPQA Diamond row).
- llm-stats.com Opus-4.7 entry (94.2 GPQA Diamond).
- VentureBeat, The Next Web, and Decrypt launch coverage (all report 94.2 vs Opus-4.6’s 91.3%).

Protocol: Anthropic’s 94.2 is maj@k with extended thinking, per the Claude 4-family methodology footnote. This is a max-effort setting. We pin 94.2 as the reported headline (the protocol-asymmetry implication is handled in §4.5 and Appendix C). Confidence: HIGH — multi-

ple independent sources, consistent value, direct vendor publication.

Generational trajectory disclosure (red-team ALT-D-2 reservation). The red-team raised an obvious question: is 94.2% an outlier / cherry-picked “peak number” from the Claude 4 family, or is it a coherent point on a generational trajectory? The available sibling-model numbers are: Claude Opus 4.5 = 87.0 GPQA Diamond, Claude Opus 4.6 = 91.3 GPQA Diamond, Claude Opus 4.7 = 94.2 GPQA Diamond (each directly published). That is a monotonically-increasing sequence of +4.3pp (4.5 → 4.6) then +2.9pp (4.6 → 4.7) — consistent with generational improvements across a typical Anthropic release cadence, without the kind of single-release jump that would suggest an outlier eval or a one-off reporting anomaly. This trajectory is the strongest single corroborator that 94.2 is not a one-off outlier. Appendix A’s provenance table carries the trajectory row verbatim.

Important caveat, also carried to Appendix A. The 94.2 value is under the Reading-B ± 3 pp band, which is explicitly “pre-declared in `sq1_benchmark_floors_v2.md` §1.3 before the gate evaluation in §4” of that file. A gate re-run under the same ± 3 pp band at a later time against a new Opus-4.7 number would potentially produce a different delta. Of particular note: at the D-109 gate re-run evaluation time, the Opus-4.7 headline 94.2 was used. One observation in the D-109 audit noted that an earlier “outlier” 94.2 trajectory datapoint FAILED the gate re-run by -43.7 pp on the closest ≤ 100 B public candidate (Llama 3.3 70B at 50.5). The trajectory 87.0 → 91.3 → 94.2 is therefore self-consistent as a published generational sequence and failure-inducing for the public open-weight frontier; both are empirically true and the paper reports both.

3) 4.3 Best ≤ 100 B FP16 public candidate on GPQA Diamond: Under the total-params reading:

- Llama 3.3 70B (dense, 70B total): 50.5% on GPQA Diamond, 0-shot CoT, single-sample, reported directly on the Meta Llama 3.3 70B model card and verified against the HuggingFace llama-3.3-70b-instruct evaluation output. This is the headline candidate score for the gate.
- Qwen2.5-72B-Instruct (dense, 72B total): approximately 60% on GPQA Diamond under Alibaba’s evaluation protocol (0-shot CoT; Qwen2.5 technical report, arXiv:2412.15115). The exact value fluctuates in the 58–62% range across reproductions; the middle estimate of 60% is the best-case eligible candidate score under total-params.
- QwQ-32B-preview (dense, 32B total): reported in the 60–65% range on GPQA Diamond under maj@k with extended-thinking protocols; the 0-shot CoT single-sample score is lower.
- DeepSeek-R1-Distill-Llama-70B: ineligible. R1-Distill models are knowledge-distilled from DeepSeek R1, which is >100 B total (671B). The distilled-student

is ≤ 100 B but carries representational imprint from a >100 B teacher, and the eligibility reading considers distillation from a non-eligible teacher as inheriting the teacher’s non-eligibility. The paper reports this explicitly because the 70B-distilled variants score 75–80 on GPQA Diamond but would under-represent the gap finding if admitted.

Under the permissive active-params reading (Appendix D):

- DeepSeek V3.2 (37B active / 671B total): $\sim 79.9\%$ on GPQA Diamond.¹ Admitted only under active-params, not under total-params.

4) 4.4 Gap magnitude, band-sensitivity, and verdict:

Under the pre-declared Reading B ± 3 pp band, the lower bound for proxy-aware-PASS on GPQA Diamond is 91.2% (94.2 – 3). The eligible candidate scores under the locked total-params reading are:

Under the active-params reading (informational, not binding): DeepSeek V3.2 at 79.9% gives delta -14.3 pp vs 94.2, and -11.1 pp vs 91.2 lower — still hard-FAIL, but the magnitude shrinks from ~ 44 pp to ~ 14 pp. Both readings therefore produce hard-FAIL verdicts; the total-params reading fails more decisively.

Band-sensitivity check. Suppose we had (counter to §8) adopted a wider band. At Reading-C-style ± 10 pp, the lower bound is 84.2; delta Llama 3.3 70B is -33.7 pp. At ± 20 pp, lower bound 74.2; delta -23.7 pp. At ± 30 pp, lower bound 64.2; delta -13.7 pp. The verdict does not flip to pass until the band is widened to ± 44 pp or the floor is relocated to ~ 52 pp. Neither is a defensible position given the 94.2% Opus-4.7 number and the three-band policy’s bands calibrated on noise-floor arguments.

5) 4.5 Aggregate amended-set verdict: `sq1_benchmark_floors_v2.md` §7 gives the summary table for the amended set under Reading B; reproduced here:

Primary hard-FAILs: 3 (MMLU, HumanEval, GPQA Diamond decisive). Primary proxy-aware-PASSes: 2 (GSM8K, TruthfulQA). Primary clean-PASSes: 0.

Secondary hard-FAIL: 1 (ARC-AGI-v2 best candidate below 4.5 lower bound under either eligibility reading).

Per Reading B aggregation (`sq1_amendment_gate_criterion_supplement.md` §2 Reading B + `sq1_amendment_addendum.md` §4 item 5), the gate passes iff (i) zero primary hard-FAILs AND (ii) secondary $\geq 5\%$. Here there are 3 primary hard-FAILs including one ~ 44 pp decisive gap, and secondary below lower bound. Overall gate verdict: FAIL.

¹DeepSeek V3.2 GPQA Diamond score pinned to primary source per C5 (red-team P5 report): DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024) reports 79.9 on GPQA Diamond for the V3 family under maj@k evaluation; the V3.2 variant inherits this evaluation per the DeepSeek-AI HuggingFace model card `deepseek-ai/DeepSeek-V3.2` (accessed 2026-04-17). Where the V3 → V3.2 GPQA Diamond value was not separately re-reported, the V3 published figure is used as the active-params candidate headline, which is conservative-against-the-paper’s-conclusion (i.e., if V3.2 were worse, the gap would widen; if better, the gap would narrow but still hard-FAIL).

Candidate	Score	Delta vs 94.2	Delta vs 91.2 lower	Verdict
Llama 3.3 70B	50.5	−43.7pp	−40.7pp	hard-FAIL (decisive)
Qwen2.5-72B	~60	−34.2pp	−31.2pp	hard-FAIL
QwQ-32B-preview	~62 (best 0-shot CoT estimate)	−32.2pp	−29.2pp	hard-FAIL

Primary floor	Opus-4.7	Band	Lower	Best $\leq 100B$	Verdict
MMLU	91.0	± 1.5	89.5	86.0 (Llama 3.3 70B)	hard-FAIL (−5.0pp vs floor; −3.5pp vs lower)
HumanEval	96.8	± 1.0	95.8	92.7 (Qwen2.5-Coder-32B)	hard-FAIL (−4.1pp vs floor; −3.1pp vs lower)
GSM8K	96.0	± 2.0	94.0	95.9 (Qwen2.5-72B)	proxy-aware-PASS (−0.1pp vs floor; +1.9pp vs lower)
GPQA Diamond	94.2	± 3.0	91.2	50.5 (Llama 3.3 70B)	hard-FAIL (−43.7pp vs floor; −40.7pp vs lower — decisive)
TruthfulQA	65.0	± 3.0	62.0	64.1 (Llama 3.3 70B median)	proxy-aware-PASS (−0.9pp vs floor; +2.1pp vs lower)
Secondary ARC-AGI-v2	≥ 5.0 threshold	± 0.5	4.5	~1.3–4 (DeepSeek R1 active-params; QwQ-32B total-params)	hard-FAIL

6) 4.6 Meta-pre-commitment fires: The meta-pre-commitment in `sql_amendment_addendum.md` §4 item 9, approved under D-109 by real PI-agent dispatch, states:

“This is the final SQ1 swap for Program 1. Any further ill-posedness under the amended set terminates Program 1 on an envelope / negative-result paper. No third-round relaxation.”

The gate FAILED under the amended set + pre-declared Reading B. The meta-pre-commitment fires. Program 1 closes on this envelope paper at ~P6. This concludes the v2 finding.

Two independent frontier falsifications. The v1 finding on ARC-AGI-v2 (§3) and the v2 finding on GPQA Diamond (§4) are independent: different benchmarks, different public-eval protocols (v2 semi-private with ARC Prize vetting; Diamond fully public with published eval scripts), different model-level scoring regimes. Both produce hard-FAIL by decisive margins. The envelope finding therefore does not rest on any one benchmark’s quirk or protocol. It rests on the public $\leq 100B$ FP16 open-weight frontier’s reach in general relative to Opus-4.7’s published floors as of 2026-04-17.

F. §5 — 18GB Envelope Implications (No-Sweep Survey)

This section is the most scope-sensitive of the paper. A reviewer who wants a sweep is not getting one here. What they are getting is a citation-dense survey of the published compression / MoE / sparsity / KV-quantization frontiers — what the literature reports as best-published results on $\leq 100B$ dense and MoE models — followed by an argument that compounding these reported results under the 18GB envelope narrows but does not close the ~44pp GPQA

Diamond gap that already exists before any envelope-side work begins.

We organize the survey into four compression regimes: inference-time weight compression (§5.1), active-parameter MoE at inference (§5.2), inference-time KV cache compression (§5.3), and training-time / distillation-time compression applied to $\leq 100B$ starting points (§5.4). §5.5 argues the compounding result. §5.6 is an explicit non-sweep caveat.

1) 5.1 Inference-time weight compression (quantization + sparsity): The 2024–2026 quantization literature has stabilized around the following empirical claims on $\leq 100B$ dense models evaluated on MMLU and GSM8K as quality proxies:

FP16 $\rightarrow$ INT8: typically 0–0.5pp MMLU drop, routinely below test-retest noise; GSM8K drop similar. Sources: GPTQ (Frantar et al.), AWQ (Lin et al.), SmoothQuant (Xiao et al.), bitsandbytes integer-8 path as implemented in HuggingFace Transformers.

FP16 $\rightarrow$ INT4 / NF4: 0.5–2pp MMLU drop at best-published settings with outlier handling (AWQ, SmoothQuant, NF4 from QLoRA), depending on model size. At 70B parameter scale the drop is typically 0.5–1pp; at 7B parameter scale it can be 1–3pp. Sources: AWQ (arXiv:2306.00978), QLoRA NF4 (arXiv:2305.14314), GPTQ (arXiv:2210.17323).

FP16 $\rightarrow$ sub-4-bit (e.g., INT2, 1.58-bit, 2-bit): 3–8pp MMLU drop at best-published settings, with outlier-handling exploratory and less established. 1.58-bit BitNet b1.58 (Ma et al., arXiv:2402.17764) reports competitive performance at scale but requires training from scratch at the reduced precision, not post-training quantization.

Sub-4-bit PTQ is an open area; reported results at 2-bit with outlier-aware schemes (GPTQ 2-bit, OmniQuant arXiv:2308.13137) typically show 4–10pp drops at $\leq 70\text{B}$ scales.

Structured sparsity on top of quantization: published combined-technique results are sparser. N:M structured sparsity (e.g., 2:4 sparsity as supported by NVIDIA Sparse Tensor Cores) on a quantized model typically costs an additional 1–2pp on MMLU under current training-time / post-training-retrainable sparsity methods (SparseGPT arXiv:2301.00774, Wanda arXiv:2306.11695). Unstructured magnitude pruning reaches higher sparsity ratios but is harder to realize as a hardware-speed gain on Apple Silicon specifically (no Sparse Tensor Core equivalent on M3 Pro).

For this program specifically: moving a Llama 3.3 70B from FP16 to INT4 would expect 0.5–1pp MMLU drop from 86.0 to ~ 85.0 – 85.5 . Stacking 2:4 sparsity adds 1–2pp; combined 2–3pp drop to ~ 83.0 – 84.0 . These losses are additive-to-subtracted in the same direction as the gap — they do not close it.

MoE-specific quantization. EAQuant (Fu et al., arXiv:2506.13329, 2025) is an expert-aware post-training quantization scheme tuned for MoE models; reports under 1pp MMLU drop at INT4 on DeepSeek-V3-scale MoE when the routing and expert heterogeneity are modeled during calibration. This is a meaningful result — but it applies to a model whose total-parameter footprint is itself disqualifying under the total-params eligibility reading. For the total-params gate, EAQuant does not open new candidates; for the active-params gate, EAQuant would additionally halve the per-expert memory footprint but still leave total storage at $\sim 84\text{GB}$ for a 671B-total model at INT4.

2) 5.2 Active-parameter MoE at inference: Recent MoE work (Mixtral 8x7B, arXiv:2401.04088; DeepSeek-V3, arXiv:2412.19437; DeepSeekMoE, arXiv:2401.06066; Sigma-MoE-Tiny, arXiv:2512.16248; GPT-OSS, arXiv:2508.10925; Hunyuan-Large, arXiv:2411.02265; Qwen3 Tech Report, arXiv:2505.09388) has established that active-parameter count at inference can be much smaller than total-parameter count, with corresponding compute savings. The joint MoE scaling laws of Ludziejewski et al. (arXiv:2502.05172, 2025), the “Greater Leverage” scaling paper (arXiv:2507.17702), and the expert-routing theoretical analysis of Wang, Hayou, and Nalisnick (Wang et al., 2026; arXiv:2604.09780, April 2026) together argue that under well-trained MoE architectures, memory-efficiency and compute-efficiency can be decoupled at inference — but also, per the Wang et al. analysis, that the specialization routing reflects feature-space geometry rather than strict domain-level specialization, which bears on expert-pruning feasibility.

For the 18GB envelope question, the decoupling is asymmetric in a specific way. Active-parameter MoE reduces compute per token (fewer FLOPs) and reduces

activation memory per token (only active experts produce activations), but does not reduce weight memory — all experts must be resident in RAM (or streamed from SSD/disk) to be routable. On a 18GB unified-memory device with no fast PCIe or NVLink bus, streaming experts from SSD introduces per-token latency of tens of milliseconds per expert layer, which is prohibitive for interactive use. Published Apple Silicon profiling (Benazir & Lin, arXiv:2508.08531, 2025) measures sustained SSD read bandwidth at approximately 2–3 GB/s on M3-class devices — a 671B-total MoE at INT4 streaming each expert once per token at $\sim 80\text{MB}/\text{expert}$ would require $\sim 80\text{MB} / 2.5\text{GB/s} \approx 32\text{ms}$ per expert activation, summing to hundreds of ms per token across multiple routed experts per layer.

The permissive “active-params $\leq 100\text{B}$ ” reading of the eligibility criterion therefore admits DeepSeek V3.2 (37B active) conceptually, but the practical envelope that such a model fits into on an 18GB device is not the standard interactive-inference regime. Expert-offload schemes (e.g., MoE-Offload and ExpertFlow, refs 30 + internal MoE-Offload literature) have published accuracy-preserving results but with 3–10 $\times$ slowdown vs dense equivalents at the active-params count. Appendix D carries the full argument.

3) 5.3 KV cache and context compression: The inference-time memory budget is not only the weight matrix; it is also the KV cache, which scales linearly in sequence length and batch size. For a 70B dense model at INT4 weights, the weight memory is approximately 35GB — already above the 18GB envelope, requiring further compression. At Q4 + 2:4 sparsity the effective weight footprint can approach $\sim 20\text{GB}$, which is marginal for 18GB; leaving an envelope of ~ -2 to 0GB for KV cache + activations + OS.

KV cache quantization. INT8 KV caches reduce memory by $\sim 2\times$ with reported 0.5–1pp MMLU drops on long-context tasks (HuggingFace integer KV implementation, HELM long-context evals). INT4 KV caches reduce by $\sim 4\times$ with 1–3pp MMLU drops depending on attention-head precision handling. KVTC (Staniszewski & Lancucki, arXiv:2511.01815, ICLR 2026) reports more aggressive KV cache transform coding achieving near-INT2-equivalent effective precision with 0.5–1pp long-context quality drop on Llama-class models at context lengths up to 32K. For the envelope question, KV quantization buys a multiplicative factor on context length more than on model size.

Context length implications. At 16GB working-set with a 70B-Q4 model at $\sim 17.5\text{GB}$ weights + $\sim 1\text{GB}$ OS/activation overhead, residual budget for KV is $\leq 0\text{GB}$ — meaning the 70B-Q4 baseline does not fit without further compression. Under 2:4 sparsity at $\sim 14\text{GB}$ weights + 2GB overhead, $\sim 2\text{GB}$ remains for KV — roughly 4K tokens of FP16 KV or $\sim 16\text{K}$ tokens at INT4 KV. This is below the $\geq 128\text{K}$ context-length regime the Opus-4.7 reference target operates at.

4) 5.4 Training-time / distillation-time compression: A from-laptop training path was part of Program 1’s original scope (SQ4: “is the SQ3 winner trainable on-laptop in ≤ 100 sessions via {scratch, distill, SFT, continual}?”). Program 1 did not reach SQ4, so we have no from-laptop training result to report. We note two published references for the envelope implications, alongside a third reference on MoE training infrastructure (Yan et al., 2026; arXiv:2603.07685, March 2026) that establishes current best-published training throughput on Megatron-Core — useful for any future from-laptop training attempt that wishes to compare its order-of-magnitude gap against published cluster training.

Knowledge distillation. Distillation from a large teacher to a ≤ 100 B student has produced notable results when the teacher is itself a frontier model: DeepSeek-R1-Distill-Llama-70B (from DeepSeek R1 teacher at 671B total) recovers $\sim 75\text{--}80\%$ of R1’s GPQA Diamond performance in a 70B dense student. This is the methodological lever that, in principle, could close the gap — if a ≤ 100 B open-weight model were distilled from Opus-4.7. As of 2026-04-17, no such distillation has been publicly reported.

Training-from-scratch at compressed precision. BitNet b1.58 (arXiv:2402.17764) reports that training from scratch at 1.58-bit ternary weights at ≥ 3 B parameter scale produces comparable or better MMLU than FP16 training at matched compute budgets. This is a positive result that, if replicated at ≤ 100 B scale, could shift the quantization curve — but it is a result about the shape of the scaling curve at compressed precision, not a claim that 1.58-bit training closes the Opus-4.7 gap. The published data are on small-scale anchors; ≤ 100 B 1.58-bit open-weight models with primary-set scores do not exist as of the paper date.

Data-constrained scaling. Muennighoff et al. (arXiv:2305.16264) establish that under data constraints (epoching effects, limited token supply), scaling behavior deviates from unlimited-data Chinchilla laws. On 18GB, data loading from a single SSD and memory-limited batch size imposes effective data budget constraints similar to the “data-repeat” regime of that paper. The implication is that from-laptop training would operate in the data-constrained regime rather than the data-unlimited regime, with corresponding sample-efficiency reductions. Not a demo, just a citation-backed statement about what the envelope would imply if one were to train from scratch on it.

5) 5.5 Compounding: why the gap does not close: Charitable compounding under best-published results on ≤ 100 B starting points (no empirical run on 18GB; this is a survey-literature compound estimate):

- Start: Llama 3.3 70B at 50.5% on GPQA Diamond (0-shot CoT). Gap vs 94.2 floor: -43.7 pp.
- + Post-training distillation from a hypothetical Opus-4.7 teacher (not publicly available): closes maybe 20–30pp on GPQA Diamond under the R1-Distill

empirical curve. Gap becomes -13.7 to -23.7 pp. Still hard-FAIL.

- + maj@k protocol equalization (see §4.5 and Appendix C): closes 5–10pp at best-published literature. Gap becomes -3.7 to -18.7 pp. Borderline to still-hard-FAIL.
- + 18GB envelope compression (Q4 + 2:4 sparsity + INT8 KV): subtracts 2–3pp from any starting point. Gap becomes -5.7 to -21.7 pp. Still hard-FAIL under Reading B ± 3 pp, and often by Reading-C ± 10 bands too.

The charitable scenario above assumes a teacher that does not exist publicly (no Opus-4.7 distillation has been released). Under the realistic scenario without such a teacher — i.e., distillation from an eligible ≤ 100 B teacher such as Qwen2.5-72B or DeepSeek-R1-Distill-Llama-70B — the ceiling of the distilled-student’s GPQA Diamond is bounded above by the teacher’s GPQA Diamond, which itself is below the floor. No iterated distillation chain has been shown to amplify a teacher’s accuracy on hard benchmarks. The realistic charitable scenario gives a gap of -30 to -40 pp even after all envelope-side compression.

Non-closure of the gap does not depend on the 18GB envelope. This is the key claim of §5. The 18GB envelope is the constraint that tightens the feasibility conclusion (smaller realizable models, tighter KV, less residual compute budget for maj@k), but the bulk of the gap is present before the envelope is applied — the frontier ≤ 100 B public open-weight starting point is 30–40pp below the Opus-4.7 Diamond floor. The envelope cannot add capability. The conclusion of §5 is therefore: under the published compression / MoE / sparsity / KV compression frontiers, no currently-feasible combination on ≤ 100 B eligible starting points closes the gap for Opus-4.7 on GPQA Diamond (or MMLU, or HumanEval, by analogous arguments).

6) 5.6 Explicit non-sweep caveat: This section did not report any from-laptop training run, any from-laptop distillation run, any from-laptop quantization-regression measurement, or any from-laptop MoE inference experiment. Every numeric claim in §5 is from a published paper or vendor disclosure and is cited. No new experimental result is claimed.

This is deliberate. The program-open memo authorizes a positive-result envelope or a negative-result envelope as equally valuable outcomes (program_open_memo.md:18–20). Running a partial sweep to be able to claim “Program 1 did X% of Opus-4.7 on Y benchmark” would be a disguised positive claim; since the gate failed decisively before any sweep, the sweep would be chasing a target the gate has already ruled out. The honest sequence is to halt at gate-FAIL and characterize the envelope from the literature, which is what §5 does.

G. §6 — §9 Aspirational-Mission Framing

This section is in plain terms the one a workshop reviewer will flag as most at risk of coming across as defeatist or as sour grapes. We have taken care to phrase it consistently with the lab’s actual values file rather than with external narrative conventions. The core claim is simple: the negative result is the anticipated function of this kind of mission, not an unexpected setback.

1) 6.1 What §9 says, in plain terms: Lab value §9 (data/values.md §9, verbatim):

“The Mission Is Aspirational On Purpose. ‘Beat Opus 4.7’ is not a tractable goal on 18GB — that’s the point. The mission exists to force the lab to develop capabilities it wouldn’t otherwise build. Reframing to an ‘achievable’ target would make the lab less ambitious, not more realistic.”

The applied rule, from the same entry:

“Director does not propose mission revisions. Milestones toward the mission (Program 1: above-random on 2 benchmarks, etc.) can be negotiated. The mission itself is fixed.”

The program-open memo then carries the corollary (program_open_memo.md:18-20):

“(a) A rigorous paper laying out the envelope with evidence + gap-closing research directions, OR (b) a negative result with specific conditions under which the mission is not currently achievable. Both are valuable. Negative with clear conditions saves years.”

2) 6.2 “Can’t do it under current techniques” is the honest answer: The operational question Program 1 set out to answer is “what fraction of Opus-4.7’s floors is achievable on 18GB?” The pre-declared falsifiability gate asks a prerequisite: is any $\leq 100\text{B}$ public open-weight starting point even within striking distance? The gate-FAIL result — decisively so on ARC-AGI-v2 at $\sim$ two-orders-of-magnitude, and independently so on GPQA Diamond at $\sim 44\text{pp}$ — is the answer to the prerequisite, and it renders the operational question ill-posed under the current public frontier.

Softening the floors to make the operational question well-posed — say, by relaxing MMLU from 91 to 86, or GPQA Diamond from 94.2 to 50, or by reframing “Opus-4.7-equivalent” to “Llama 3.3 70B-equivalent” — would produce a program where the answer is trivially “yes, we can fit Llama 3.3 70B-equivalent.” That answer is correct but uninteresting. It is also, in §9 terms, a less ambitious question than the one the lab opened. §9 is explicit that milestones along the path (e.g., “Program 1 gets above-random on 2 benchmarks”) are negotiable; the mission itself is not. The envelope paper therefore does not argue for mission revision. It reports on the current state of the public frontier + envelope conjunction and marks the mission as “not achievable under current techniques,” which §9 anticipates as a valid outcome.

3) 6.3 Reading C as a case study in pre-commitment discipline: The Reading C decision point in §4 is the single place in the program where the lab could have produced a “yes” answer to the operational question at the cost of discipline. The parent amendment proposal’s “Expected outcome: pass (within 10-20pp)” footnote would, if applied as a gate criterion, have produced a “proxy-aware-PASS” verdict for Llama 3.3 70B on GPQA Diamond (50.5 at ± 10 lower 84.2 $\rightarrow$ hard-FAIL but borderline; at ± 20 lower 74.2 $\rightarrow$ hard-FAIL but closer; at ± 30 lower 64.2 $\rightarrow$ still hard-FAIL on Llama 3.3 but proxy-aware-PASS on Qwen2.5-72B at 60; at ± 34 lower 60 $\rightarrow$ proxy-aware-PASS on Qwen2.5-72B).

The Director’s self-critique supplement surfaced this ambiguity before the PI signed. The PI explicitly rejected Reading C on §8 grounds — not because the ± 10 -20pp band was wrong in some absolute sense, but because it was nowhere pre-declared as a gate criterion; adopting it silently at gate time would have been post-hoc criterion-shopping. The PI selected Reading B as the minimum-consistent extension of the already-locked three-band policy. The decision record (data/agents/pi/episodic/2026-04-17_program_1_phase_1_and_2.md Part B) is preserved as an audit artifact.

The value of this decision record for the paper is not “our PI is strict” (that is organizational); it is that a feasibility-envelope negative result depends on pre-commitment discipline to be scientifically meaningful. A gate that silently widens to accommodate the data is not a gate. The envelope finding rests on the bands not being widened, which required an explicit opportunity to widen them and an explicit refusal. The refusal is as much a methodological contribution of the paper as the floor numbers.

4) 6.4 What the honest answer unlocks: A well-characterized negative result with specific conditions is a forward pointer. The conditions under which the Program 1 verdict would flip are enumerated in sq1_benchmark_floors_v2.md §6.5 and paraphrased here:

A new $\leq 100\text{B}$ FP16 open-weight model release that clears MMLU ≥ 89.5 AND HumanEval ≥ 95.8 AND GSM8K ≥ 94.0 AND GPQA Diamond ≥ 91.2 AND TruthfulQA ≥ 62.0 AND ARC-AGI-v2 $\geq 5\%$ simultaneously would reset the gate result. Such a model would make Program 1’s operational question tractable-in-principle, and a follow-up envelope measurement on that starting point would be the right next step.

As of 2026-04-17 no such model is publicly available. The closest-in-class release we are aware of is Qwen3.5-235B-A22B (235B total, 22B active), which exceeds the 100B total threshold and is ineligible under the locked reading. Absent a threshold-eligible release with the required floor-clearing behavior, the envelope finding of this paper stands. A future such release would reset the question in a new program open, not in an amendment to

this one (the meta-pre-commitment binds this program, not the successor).

5) 6.5 The institutional point: The paper reports the refusal-to-relax as part of the methodology, not as an aside. This is deliberate: the lab’s pre-commitment discipline is the load-bearing institutional property that makes the falsifiability gate meaningful. A lab without such discipline, running a similar program, would have either (a) not built the gate into SQ1, producing a different failure mode (gate-less “partial positive” claim), or (b) silently relaxed the gate at run time, producing an unfalsifiable “yes” answer. We report the discipline not as virtue-signaling but as a technical prerequisite for the result to mean what it says.

H. §7 — Methodology (Body Summary; Detail in Appendices)

This section lists the four methodological contributions of Program 1 that outlive the negative result. Full technical detail lives in the appendices; this section summarizes and links.

1) 7.1 Three-band proxy-aware falsifiability policy (Appendix B): The three-band policy (p1_4_proxy_acknowledgment.md §3; Appendix B of this paper) replaces a hard-threshold accept/reject policy with three tiers (hard-FAIL / proxy-aware-PASS / clean-PASS) with per-benchmark pre-declared band widths motivated by (i) saturation effects at each benchmark’s current ceiling, (ii) harness variance across lm-eval / HELM / HuggingFace OpenLLM / vendor re-implementations, and (iii) standard error of sampling proportion at Diamond-class evaluation set sizes.

The policy applies symmetrically to proxy floors (MMLU, GSM8K, TruthfulQA inherited from sibling models) and to direct floors (HumanEval third-party reported, GPQA Diamond direct Anthropic publication). Reading B (§4.1) is the name we use for extending the three-band policy to the SQ1 falsifiability gate itself rather than only to downstream SQ2–SQ5 candidate evaluation.

Contribution beyond Program 1. Any capability-envelope program operating against a public-model frontier faces the same proxy + band issue: public-model scores are noisy across protocols, reference floors often have to be proxy-inherited from sibling models, and hard 1pp thresholds produce unjustified pass/fail flips on borderline cases. The three-band policy is generalizable; a follow-up methods paper on the policy specifically, with broader empirical calibration of the per-benchmark bands, is a feasible separate publication. The policy is presented in Appendix B in a self-contained form to support such reuse.

2) 7.2 Protocol-asymmetry corrections (Appendix C): Opus-4.7’s published scores are routinely under max-effort protocols (maj@k + extended thinking), while $\leq 100\text{B}$ open-weight scores are typically 0-shot CoT single-pass. A naive comparison without protocol-equalization would systematically underestimate the open-weight side. Appendix

C documents the empirical literature on per-benchmark maj@k gains:

- GPQA Diamond: maj@k gains in the 5–10pp range for k up to 64, per published reasoning-model papers.
- MMLU: maj@k gains 1–3pp; MMLU is largely knowledge-retrieval, not sampling-benefit dominant.
- HumanEval: maj@k gains $< 1\text{pp}$ at the 96%+ ceiling; near-saturation makes sampling gains marginal.
- GSM8K: maj@k gains 2–4pp typical at the 90%+ level; saturation applies similarly to MMLU.
- TruthfulQA MC2: harness variance 2–4pp across Eleuther / HELM / lm-eval-harness implementations dominates maj@k effects.
- ARC-AGI-v2: private-eval protocol dominates; maj@k benefits unreported publicly.

Decision rule the paper adopted. When a gap between Opus-4.7 floor and best $\leq 100\text{B}$ candidate is larger than twice the per-benchmark maj@k ceiling gain (e.g., GPQA Diamond gap $\geq 20\text{pp}$), protocol-equalization is not load-bearing for the verdict; the verdict is robust to any plausible protocol adjustment. When the gap is within $\leq 2\times$ the maj@k ceiling gain, protocol-equalization becomes load-bearing and the gate reports a protocol-equalized number alongside the headline. For the amended set, only GSM8K and TruthfulQA are in the protocol-sensitive range; MMLU, HumanEval, and especially GPQA Diamond are not.

3) 7.3 MoE total-vs-active eligibility argument (Appendix D): The locked eligibility reading of $\leq 100\text{B}$ total parameters is defended against the permissive active-params $\leq 100\text{B}$ reading. Appendix D carries the full argument. The short version:

- Total-params reading wins on storage. The binding 18GB constraint is RAM-resident model size, not compute per token. A 671B-total MoE at Q4 needs $\sim 336\text{GB}$ storage regardless of active-params routing. No on-device inference instantiation with standard interactive latency is possible.
- Active-params reading wins on FLOPs. Per-token FLOP budget scales with active params. A 37B-active MoE has inference-compute parity with a 37B dense. On a compute-budgeted device this matters.
- Disambiguation at the 18GB envelope. The laptop is memory-budgeted, not compute-budgeted (the M3 Pro GPU has substantial FLOP headroom relative to what an 18GB-fitting model can use). Total-params reading is the correct eligibility criterion.
- Under both readings the gate fails. Gate fails on GPQA Diamond at -14.3pp active-params (DeepSeek V3.2 at 79.9^2 vs 91.2 lower) and -43.7pp total-params (Llama 3.3 70B at 50.5 vs 91.2 lower). Not a reading-dependent result.

²DeepSeek V3.2 79.9 GPQA Diamond primary source: DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024); HuggingFace model card deepseek-ai/DeepSeek-V3.2 (accessed 2026-04-17). See footnote at §4.3 for full provenance discussion.

4) 7.4 Floor-provenance audit and re-baseline protocol (Appendix A): The paper’s claims rest entirely on external numeric floors (Opus-4.7’s published scores). A provenance audit is therefore a methodological prerequisite, not a stylistic nicety. Appendix A tabulates each floor with source type (direct Anthropic / range-published / third-party / proxy-inherited), primary source + URL + accessed date, corroborating sources, protocol, confidence, and proxy error-bar. The audit is most load-bearing on the GPQA Diamond row, where the generational trajectory disclosure (Opus 4.5 = 87.0 $\rightarrow$ 4.6 = 91.3 $\rightarrow$ 4.7 = 94.2) is the strongest single corroborator of the 94.2% Opus-4.7 value against an outlier-eval hypothesis.

The audit carries a re-baseline protocol (p1_4_proxy_acknowledgment.md §3.2): on authoritative publication of a new Anthropic-disclosed number for any floor, the floor is auto-updated in a dated errata entry to the companion open-data artifact (sq1_benchmark_floors_v2.md), with the superseded number preserved under §4 archive-never-delete. If the aggregate floor shift across all primary floors exceeds 5pp (a kill-criterion threshold added to question.md per the amendment), Phase 2 is re-opened. No such trigger has fired as of the paper date.

5) 7.5 Pre-commitment as a methodology, not a virtue: Finally, the meta-pre-commitment (sq1_amendment_addendum.md §4 item 9 — “this is the final SQ1 swap; next FAIL $\rightarrow$ paper”) is itself a methodological contribution. It generalizes beyond Program 1 as the following pattern: for any falsifiability-driven capability-envelope program, at the moment of an authorized one-shot relaxation, attach a binding commitment that the relaxation is final and that any second failure terminates the program. This pattern forecloses the post-hoc goalpost-movement failure mode that §8 pre-commitment exists to prevent, without foreclosing the single relaxation that a good-faith amendment of an ill-posed set requires. It is a specific technical shape of the §8 abstract principle. The paper reports it as such.

I. §8 — Conclusion + Forward Pointers

1) 8.1 Closure claim (one sentence): Under the locked eligibility criterion ($\leq 100B$ FP16 publicly-released open weights, total parameters), evaluated against the post-amendment primary benchmark set {MMLU, HumanEval, GSM8K, GPQA Diamond, TruthfulQA} with the pre-declared three-band proxy-aware policy (Reading B) and hard secondary floor (ARC-AGI-v2 $\geq 5\%$), no existing configuration clears the Opus-4.7 floor vector; consequently, no 18GB laptop-side inference-only path and no from-laptop-trained/distilled path yields an Opus-4.7-equivalent instantiation under current compression / MoE / sparsity techniques and the public open-weight frontier as of 2026-04-17.

2) 8.2 What this closes:

- Program 1’s scope, as defined in program_open_memo.md. The program was an aspirational feasibility

question about laptop-scale frontier-equivalence under 2026-vintage techniques + public frontier. The question is answered in the negative; closure is correct.

- SQ1 locks, as updated in sq1_benchmark_floors_v2.md + unanimous_lock_li.md. No re-open pathway exists within Program 1.
- SQ2 through SQ5 are not revisited. They were blocked at gate-FAIL and remain so.

3) 8.3 What this does NOT re-open:

- question.md (locked). The operational question as-written is what was falsified; rewriting the question post-falsification would be a §8 violation at the program level (beyond the sub-question level).
- The five-benchmark set (meta-pre-committed). Any further relaxation requires a new program-open memo, not an SQ1 amendment.
- The eligibility criterion (pre-committed). Total parameters $\leq 100B$ FP16 stands.
- The three-band policy (locked via D-109 PI AP-PROVE). Reading B stands.

4) 8.4 Forward pointers (informational only):

- Gap-closure vector. A new $\leq 100B$ FP16 open-weight release that clears the full amended primary set + secondary would reset the question. Tracking is a future-Program conversation, not an SQ1 amendment.
- Methodology carry. The three-band policy + meta-pre-commitment pattern generalizes beyond Program 1. A follow-up methods paper on the policy is feasible.
- Apparatus carry. No apparatus was built in Program 1 past Phase 2 (P7+ skipped). The lab’s infrastructure (Makefile build system, dashboard, hardware monitor, experiment tracker) was not exercised on a from-laptop run in this program but is preserved for Program 2+. The infrastructure is independent of the negative result and remains usable.
- Red-team pattern carry. The adversarial review pattern that produced Attack 3 and Attack 4 (promoting diagnostics to hard secondary floors; attaching a meta-pre-commitment to any one-shot relaxation) is generalizable to future programs. The data/engineering/red_team_amendment_review_2026-04-17.md record is preserved as a worked example.

5) 8.5 Explicit refusal of pivot framings: This paper does not reframe as “Program 1-lite achieves X% of Opus-4.7 on some subset of floors under some subset of techniques.” No such experiment was run. The negative result is clean precisely because no partial positive claim is smuggled in. A reviewer or a future Director session that believes a partial positive claim would improve the paper should propose a new program (with its own open-memo, sub-questions, and falsifiability gates) rather than an edit to this one.

6) 8.6 Closing note: The mission stands. §9 anticipates this category of outcome, and the closure of Program 1 is a specific condition under which the mission is not currently achievable — not a revision of the mission itself. The next program, whenever it opens, inherits the mission and inherits the carry-over from this paper’s methodology + provenance audit. That is the path forward.

J. Appendix A — Floor Provenance Table (ALT-D-2 Carry-Forward)

This appendix pre-addresses the red-team critique that Opus-4.7’s 94.2% GPQA Diamond floor could be an outlier / cherry-picked result (ALT-D-2 reservation from data/state.md §Carry-Forward Flags; red-team findings data/engineering/red_team_amendment_review_-2026-04-17.md Attack 2 Director disposition). Each row lists the floor, Opus-4.7 value, source type, primary source + accessed date, corroborating sources, protocol, confidence, and proxy error-bar. Row 4 (GPQA Diamond) inlines primary-source URL + access date directly in the table cell per C4 of the P5 red-team report; other row primary-source URLs are preserved in the companion artifact `sql_benchmark_floors_v2.md` (CC-BY-4.0) and accessible for mechanical re-execution.

1) A.1 Provenance table:

2) A.2 GPQA Diamond trajectory disclosure (ALT-D-2 critical row): Trajectory: Opus 4.5 = 87.0 → Opus 4.6 = 91.3 → Opus 4.7 = 94.2 on GPQA Diamond. Each value is directly published by Anthropic in the respective launch materials. The progression shows +4.3pp (4.5 → 4.6) and +2.9pp (4.6 → 4.7), consistent with typical Anthropic generation-to-generation improvement on the Claude 4 family (as observed across MMLU, GSM8K, HumanEval for the same siblings). No single-release jump exceeds the typical cadence. The trajectory is the strongest single corroborator against an “outlier-eval” hypothesis for the 94.2 value: an outlier or cherry-picked benchmark run would typically not sit on a monotonic sequence but would appear as an anomalous spike relative to siblings.

Note on outlier-trajectory failure in the gate re-run. The gate re-run reported in §4 of this paper used the Opus-4.7 headline 94.2 as the GPQA Diamond floor. The Reading B band ± 3 pp was pre-declared in `sql_benchmark_floors_v2.md` §1.3 before the gate evaluation. The closest eligible public candidate (Llama 3.3 70B at 50.5) FAILED the gate re-run with delta -43.7 pp from the 94.2 floor and -40.7 pp from the 91.2 Reading-B lower bound. The decisive-FAIL verdict is what closed Program 1. The trajectory disclosure itself is corroborative of 94.2 as the reliable Opus-4.7 floor value, not a second independent data point against the frontier — the gap finding rests on Llama 3.3 50.5 vs 94.2, not on Llama 3.3 vs 87.0/91.3/94.2 across time.

3) A.3 Protocol caveat: The 94.2 GPQA Diamond value is reported by Anthropic under maj@k with extended thinking. A 0-shot CoT single-sample Opus-4.7 Diamond

run would be $\sim 5\text{--}10$ pp lower per the literature priors (§7.2 / Appendix C). Even after maximally generous protocol equalization (subtract 10pp from 94.2 to get a hypothetical 0-shot single-sample Opus-4.7 Diamond floor of 84.2), the gap to Llama 3.3 70B 0-shot single-sample Diamond of 50.5 remains 33.7pp. Protocol equalization does not close the verdict.

4) A.4 Re-baseline trigger record: As of 2026-04-17, no floor has been re-baselined post-lock. The companion artifact `sql_benchmark_floors_v2.md` §6.4 documents the protocol: authoritative new Anthropic publication triggers automatic re-baseline with dated errata, and aggregate shifts ≥ 5 pp across floors re-open Phase 2. Neither trigger has fired. Any future trigger produces an errata entry on the companion artifact, not a retraction of this paper (§4 archive-never-delete).

K. Appendix B — Three-Band Proxy-Aware Policy (Full Specification)

Compliance note: this appendix is kept under 1500 words per the dispatch scope bound.

1) B.1 Formal definitions: Let f be the Opus-4.7 floor value (in pp) for a given benchmark, b be the benchmark’s pre-declared band width, and s be a candidate model’s observed score on the benchmark under the specified evaluation protocol. Define:

- clean-PASS: $s \geq f$. The candidate is at or above the floor.
- proxy-aware-PASS: $f - b \leq s < f$. The candidate is below the floor but within the per-benchmark band — a zone where proxy-inheritance uncertainty, harness variance, or protocol asymmetry plausibly explains the gap.
- hard-FAIL: $s < f - b$. The candidate is beyond the band; the gap is not attributable to proxy or noise.

2) B.2 Per-benchmark band derivation: Per-benchmark band widths (`p1_4_proxy_acknowledgment.md` §3.1; `sql_benchmark_floors_v2.md` §1.3):

- MMLU ± 1.5 pp. Motivation: MMLU 5-shot is near-saturated at Opus-4.7 floor 91.0; absolute test-retest noise at this ceiling is 0.5–1.0pp per published reproductions. The 1.5pp band adds 0.5–1.0pp of proxy-inheritance uncertainty (Opus-4.7 MMLU is extrapolated from 4.5/4.6 disclosure ranges). Sum: 1.5pp.
- HumanEval ± 1.0 pp. Motivation: near-ceiling at 96.8; pass@1 single-sample noise is < 0.5 pp; third-party reporting (not direct Anthropic) adds 0.5pp uncertainty. Sum: 1.0pp.
- GSM8K ± 2.0 pp. Motivation: saturation at 96.0 with CoT protocol variance contributing 1.0–1.5pp across lm-eval / HuggingFace / HELM reproductions; proxy-from-saturation adds 0.5–1.0pp. Sum: 2.0pp.
- GPQA Diamond ± 3.0 pp. Motivation: three independent components. (i) The Diamond subset N=198

produces standard error of sampling proportion at mid-range accuracy of $\sqrt{p(1-p)/N} \approx 3.5\text{pp}$ at $p=0.5$, i.e., roughly 1σ binomial sampling noise. (ii) Published model-to-model variation at 70–90% accuracy on Diamond is 1–4pp across independent evaluations. (iii) Protocol asymmetry (Opus maj@k vs open-weight 0-shot CoT) adds 5–10pp expected maj@k gain, but the $\pm 3\text{pp}$ band does NOT absorb this — protocol asymmetry is handled separately by the decision rule in Appendix C, not by widening the band to hide the gap. The $\pm 3\text{pp}$ band is therefore a conservative-but-defensible noise floor consistent with TruthfulQA MC2 (also $\pm 3\text{pp}$).

- TruthfulQA MC2 $\pm 3.0\text{pp}$. Motivation: harness variance 2–4pp across Eleuther lm-eval / HELM / lm-eval-harness implementations is the dominant source. Proxy-inheritance from GPT-4-family extrapolation adds $<1\text{pp}$ relative to the dominant harness noise. Sum: 3.0pp.
- ARC-AGI-v2 $\pm 0.5\text{pp}$. Motivation: ARC Prize semi-private evaluation has low test-retest noise (vetted submissions are reproducible within 0.5pp on public-eval). The $\pm 0.5\text{pp}$ band is the practical eval-set noise floor.

3) B.3 Rule C binding (evaluator discipline): The evaluator (per p1_4_proxy_acknowledgment.md §2.2 Rule C) subordinates point scores to confidence intervals. In practice:

- Opus-4.7 floors are treated as point estimates with the per-benchmark bands reflecting proxy/harness/protocol uncertainty.
- Candidate scores are reported with observed-value primacy; when a candidate score varies across sources (e.g., Llama 3.3 70B MMLU reported 86.0–86.5 across HuggingFace + Meta card), the evaluator uses the observed range without cherry-picking the favorable end.
- Borderline cases (proxy-aware-PASS near-hard-FAIL boundary) are not tipped by single-source scores; corroboration from ≥ 2 independent sources is required.

4) B.4 Aggregation under Reading B: Primary-set aggregation:

- Gate PASS iff zero primary floors hard-FAIL under Reading B bands.
- Gate FAIL iff ≥ 1 primary floor hard-FAILs.
- Proxy-aware-PASSes are counted toward gate PASS; clean-PASSes strictly dominate.

Secondary-set aggregation:

- Secondary PASS iff ARC-AGI-v2 best-eligible $\geq 5\% - 0.5\text{pp} = 4.5\%$.
- Secondary FAIL independent of primary (i.e., both must pass simultaneously).

Overall gate verdict: PASS iff primary PASS AND secondary PASS.

5) B.5 Re-baseline protocol: On authoritative publication by Anthropic of a new floor value for any benchmark:

1. Companion artifact `sql_benchmark_floors_v2.md` is updated with a dated errata entry; the superseded value is preserved per §4 archive-never-delete.
2. Aggregate floor shift across all primary floors is computed. If $\Sigma|\Delta_i| \geq 5\text{pp}$, Phase 2 is re-opened in a new program (not an amendment — the meta-pre-commitment binds this program).
3. The paper’s envelope finding is not retracted. An errata entry points to the new companion snapshot.
- 6) B.6 Rationale for rejecting Readings A and C:

- Reading A (strict original gate, no bands): pre-dates the three-band policy; Reading B subsumes it cleanly and is the minimum-consistent extension. Accepting A would mean applying a hard-threshold policy to the gate while the rest of the program uses bands — internal inconsistency.
- Reading C ($\pm 10\text{--}20\text{pp}$): nowhere pre-declared; appears only as an “expected outcome” footnote in the parent amendment proposal. Adopting it silently at gate time is post-hoc criterion-shopping regardless of signature. §8 pre-commitment violation. The PI explicitly rejected Reading C on these grounds (D-109 appendix).

7) B.7 Generalization: The three-band policy applies generally to capability-envelope programs where (i) reference floors are proxy-inherited, (ii) evaluation protocols differ between reference and candidate, and (iii) benchmark noise floors vary across benchmarks. The per-benchmark band widths must be recalibrated for the specific benchmark mix of the new program; the policy structure is program-independent.

L. Appendix C — Protocol-Asymmetry Corrections

1) C.1 The asymmetry: Vendor-published scores (Opus-4.7 in particular) are often max-effort: majority-vote over k samples (maj@ k), extended thinking with deliberation-scale compute, best-of- n sampling, or other test-time inference enhancements. Open-weight $\leq 100\text{B}$ model scores are typically 0-shot CoT with single-sample decoding. A naive comparison would systematically disadvantage the open-weight side.

2) C.2 Per-benchmark maj@ k gain literature:

- GPQA Diamond. Best-published maj@ k gains on GPQA Diamond span 5–10pp for k up to 64 on $\leq 100\text{B}$ reasoning-trained models (per the Qwen QwQ, DeepSeek R1, and o1-style literature). At $k > 64$, gains plateau. The 5–10pp range is the appropriate “protocol-equalization budget” for Opus-4.7 Diamond.
- MMLU. Maj@ k gains 1–3pp on knowledge-retrieval questions; MMLU is largely recall, less sensitive to sampling-benefit than reasoning benchmarks.

- HumanEval. Maj@k gains <1pp at the 96%+ saturation ceiling; most pass@1 failures are genuine model capability gaps, not sampling variance.
- GSM8K. Maj@k gains 2–4pp typical at the 90%+ level; saturation dampens gains similarly to MMLU.
- TruthfulQA MC2. Harness variance (Eleuther lm-eval vs HELM vs lm-eval-harness) contributes 2–4pp and dominates sampling-protocol effects; maj@k gains unclear.
- ARC-AGI-v2. ARC Prize evaluation protocol is vendor-specific; maj@k benefits are not publicly reported in a comparable form.

3) C.3 Decision rule for when protocol-equalization is load-bearing: Rule: protocol-equalization is load-bearing (i.e., potentially flips the verdict) iff the observed Opus-vs-candidate gap is within $2 \times$ (per-benchmark maj@k ceiling gain). Otherwise, the verdict is robust to any plausible protocol adjustment and the gap can be reported as-is.

Applied to the amended-set results:

- MMLU 91.0 – 86.0 = 5.0pp gap. $2 \times$ max gain = 6pp. Load-bearing — protocol-equalization could flip borderline result. But observed gap 5pp is already near-gap and the verdict under Reading B ± 1.5 pp is hard-FAIL regardless of up to 3pp maj@k credit to candidate.
- HumanEval 96.8 – 92.7 = 4.1pp gap. $2 \times$ max gain = 2pp. Not load-bearing — verdict robust.
- GSM8K 96.0 – 95.9 = 0.1pp gap. Far within $2 \times$ max gain of 8pp. Load-bearing — but verdict is proxy-aware-PASS either way.
- GPQA Diamond 94.2 – 50.5 = 43.7pp gap. $2 \times$ max gain = 20pp. Not load-bearing — verdict robust by ~ 24 pp margin.
- TruthfulQA 65.0 – 64.1 = 0.9pp gap. Well within $2 \times$ max gain. Load-bearing — but verdict proxy-aware-PASS either way.

4) C.4 Aggregate impact: For the amended-set gate specifically, protocol-equalization is load-bearing on MMLU (where the gap is within $2 \times$ max gain), non-load-bearing on HumanEval and GPQA Diamond (gaps are large relative to max gain), and not relevant on GSM8K and TruthfulQA (verdicts are pass either way). Crucially the decisive GPQA Diamond hard-FAIL by ~ 44 pp is robust to any realistic protocol equalization; the verdict does not flip. Rectifying MMLU with protocol-equalization moves Llama 3.3 from 86 to ~ 87 –88 at most, which remains below the 89.5 Reading-B lower bound.

5) C.5 Saturation effects: At benchmark ceilings >95%, the mechanical floor of 1pp absolute shifts starts to cross test-retest noise. For GSM8K at 96.0 and HumanEval at 96.8, any observed candidate score within 1–2pp of ceiling is epistemically within noise. This is part of why the three-band policy tightens the band on saturated benchmarks (± 2.0 GSM8K, ± 1.0 HumanEval) rather than loosening.

6) C.6 When protocol-equalization crosses into criterion relaxation: There is a line between protocol-equalization

(legitimate methodological correction) and criterion relaxation (illegitimate goalpost movement). The three-band policy places this line at the band width: adjustments within the band are allowed (proxy-aware-PASS already absorbs them); adjustments beyond the band are not allowed because they would re-introduce post-hoc shopping. Reading B honors this line; Reading C (hypothetically ± 20 pp) would cross it.

M. Appendix D — MoE Total-vs-Active Eligibility Reasoning

1) D.1 The argument for total-parameters ≤ 100 B: Storage is the binding 18GB constraint on the M3 Pro laptop. A model must be RAM-resident (or streamable from SSD without prohibitive per-token latency) to be runnable. A 671B-total MoE at Q4 occupies ~ 336 GB; at FP16 it occupies ~ 1342 GB. Neither fits the 18GB physical pool, nor the 119GB local SSD, nor the ~ 1.3 TB iCloud Drive at any reasonable streaming-rate under the M3 Pro SSD bandwidth limit (sustained ~ 2 –3 GB/s per Benazir & Lin, arXiv:2508.08531).

Running a 671B-total MoE with expert offloading would require per-token expert-fetch latencies of tens to hundreds of milliseconds at Q4 expert sizes of ~ 80 MB each, summed across the routed expert count per layer and the layer count. This is an order of magnitude slower than interactive-inference latency budgets. On-device inference with standard interactive UX for 671B-total MoE models is not feasible on 18GB M3 Pro hardware under current techniques.

Therefore, total-parameters is the correct eligibility criterion for the 18GB envelope question. The locked criterion in subquestions `lg.md` and `sql_benchmark_floors.md` uses total-parameters throughout, consistent with this argument.

2) D.2 The steelman for active-parameters ≤ 100 B: The counterargument: FLOP budget scales with active parameters, not total parameters. A 37B-active MoE (e.g., DeepSeek V3.2) has inference-compute parity with a 37B dense on a per-token basis. On a compute-budgeted device (e.g., a high-end GPU with plenty of RAM), active-params is the right efficiency metric.

Further: expert-offload schemes such as MoE-Offload and ExpertFlow (ref 30) have published accuracy-preserving results with 3 – $10 \times$ slowdown vs dense equivalents. If “interactive” latency budgets are relaxed, or if the use case is batch/offline inference, the active-params reading could be defensible even on 18GB via aggressive expert offload + bandwidth optimization.

Finally: the joint MoE scaling laws of Ludziejewski et al. (arXiv:2502.05172) argue that memory-efficiency and compute-efficiency can be decoupled in well-trained MoE architectures, suggesting that for scientific-question purposes the “compute-per-capability” metric (active-params-based) is the meaningful efficiency dimension.

3) D.3 Disambiguation at the 18GB envelope: The 18GB M3 Pro laptop is memory-budgeted, not compute-budgeted. The M3 Pro GPU has substantial FLOP headroom relative to what an 18GB-fitting model can use (roughly 2–4× FLOP headroom on a 70B-Q4 model based on published M3 Pro benchmarks). The compute-efficiency argument for active-params does not apply to this device class. Storage is the binding constraint.

In addition: the stated use case in the program’s open memo is interactive laptop-side inference (not batch / offline), which rules out the relaxed-latency regime where expert offload would be viable.

Therefore, under the specific device + use-case the program asks about, total-params is the correct eligibility reading. The paper locks this reading.

4) D.4 Results under both readings: We run the gate under both readings for transparency:

- Under total-params $\leq 100\text{B}$: eligible candidates are Llama 3.3 70B, Qwen2.5-72B, QwQ-32B-preview, smaller Mistral/Llama/DeepSeek $\leq 100\text{B}$ dense, etc. Best GPQA Diamond = Llama 3.3 70B at 50.5. Gate fails at -43.7pp .
- Under active-params $\leq 100\text{B}$: additionally-eligible candidates include DeepSeek V3.2 (37B active / 671B total), Qwen3.5-235B-A22B (22B active), Mixtral-8x22B variants. Best GPQA Diamond = DeepSeek V3.2 at ~ 79.9 .³ Gate fails at -14.3pp .

Both readings fail. The total-params reading fails decisively ($\sim 44\text{pp}$); the active-params reading fails robustly ($\sim 14\text{pp}$, still $>4\times$ the Reading B $\pm 3\text{pp}$ band). No choice of eligibility reading produces a PASS verdict on the amended primary set.

5) D.5 Recommendation for future programs: If expert-offload + fast-SSD streaming matures to within $2\times$ dense-model latency on M3-class hardware, a future program should re-open the active-params framing. The analysis is beyond this program’s scope. Such a future program opens with its own program-open memo, its own falsifiability gate, and its own eligibility reading; it does not retroactively amend Program 1.

N. Appendix E — Publication Decision Stub (P6 Deliverable, v2 Expanded)

This appendix is the v2 expansion of the v1 Appendix E in response to C1 of the P5 red-team report. The v1 version deferred the authorship-convention and venue-fallback questions to P6 as “open questions.” The v2 version (this one) locks the defaults that are within paper-writer scope and makes the escalation path explicit for anything requiring Director or PI decision.

³DeepSeek V3.2 GPQA Diamond 79.9 primary source: DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024); HuggingFace model card deepseek-ai/DeepSeek-V3.2 (accessed 2026-04-17). See footnote at §4.3 for full discussion including the V3 $\rightarrow$ V3.2 inheritance argument.

1) E.1 Venue-fallback table (v2 expansion per C1): The table below maps candidate venues to their authorship policy, the lab’s operational path under that policy, and the consequence of any venue rejecting the locked collective-authorship convention (option (a) in §E.2).

Key takeaway from the table: the primary path for Program 1’s Alt-D envelope paper is the workshop tier + arXiv + internal archive, which together form a publication-complete triple independent of any main-conference decision. The aspirational tier is intentionally deferred and is not part of Program 1’s P6 close; if a future Director + PI decision wants to pursue main-track submission, that is a separate future-program-level decision requiring human-author recruitment.

Robust fallback principle (explicitly stated per C1 red-team recommendation): if any external venue rejects the collective-authorship format, the lab’s fallback is the arXiv + internal archive path, NOT the silent introduction of a figurehead human author onto the paper. Silently crediting a human who did not contribute to the work would be a publication-ethics violation and violates the lab’s D-109 governance standard. The arXiv + internal archive path is publication-complete on its own terms.

Submitter-of-record convention. For any external-facing submission, the submitter of record (the human account-holder who must exist by administrative requirement of arXiv, workshop submission systems, and similar) is Harshith Kantamneni (hkantamneni2@wisc.edu, University of Wisconsin–Madison). This is a conventional administrative role — the submitter signs ethics statements, receives correspondence, and assents to the venue’s publication terms. It does NOT confer authorship credit under the lab’s convention; authorship remains “AGI Lab (autonomous agent collective)” with agent-role contributor statements. This separation is analogous to a PI-as-submitter convention in labs where the student-authors do the work and the PI submits on their behalf.

2) E.2 Authorship convention (v2 lock at option (a)): Locked convention: option (a) — collective authorship with agent-role credits in contributor statement.

Authorship line (template, applied at paper header): “AGI Lab (autonomous agent collective), Program 1, Alt-D branch.”

Contributor statement (template, applied at paper header): “Contributors of record: chief_scientist (outline + methodology), paper_writer (draft), measurement_theorist (floor pinning + gate re-run), red_team (adversarial review), evaluator (closure verdict), director (dispatch + synthesis), pi (co-sign + §9 framing). Per D-109, all agent roles — including PI — are instances of Anthropic’s Claude Opus 4.7 operating under role-specific semantic and procedural prompts. No external human contributed content.”

Submitter-of-record line (template, applied at paper header when required by venue): “Submitter of record (external-facing venues): Harshith Kantamneni, hkantam-

neni2@wisc.edu, University of Wisconsin–Madison. Administrative submitter under the venue’s account-holder requirement; does not confer authorship credit.”

Why option (a) is locked here rather than deferred to P6. The P5 red-team report C1 flagged that v1’s “open convention question” framing left too much uncertainty about the paper’s publishable form. Paper-writer has authority over authorship-convention decisions as they apply to the paper’s header (this is paper-writer scope per the procedural file); Director + PI retain authority over lab-wide authorship convention policy (which is a governance question, beyond this paper). The v2 lock applies option (a) to this paper specifically under paper-writer scope. If a future Director + PI governance decision standardizes a different lab-wide convention (e.g., option (b) individual agent author-lines, or option (c) human-equivalent-only authorship with agent-role acknowledgment), this paper can be re-emitted under the new convention at the P6 close.

Why options (b) and (c) were not adopted for this paper. - Option (b) (individual agent author-lines) would list each agent role as a separate “author” (e.g., Author 1: chief_scientist, Author 2: paper_writer, ...). This conflicts with existing authorship ethics literature that assumes authors are individuals with durable identities; agent roles are semantic-memory-bound instances of the same underlying model (Claude Opus 4.7) and do not have durable identities outside the lab’s memory files. Option (b) risks misrepresentation. - Option (c) (human-equivalent-only authorship with agent-role acknowledgment) would list a human (e.g., the submitter-of-record) as the sole author and relegate agent roles to an acknowledgment section. This conflicts with D-109 which recognizes agent roles as contributors, not as tools. Option (c) risks under-crediting the agent collective.

Option (a) — collective authorship with agent-role credits — is the minimum-conflict convention: it credits the collective that actually did the work (aligning with D-109), and it avoids the misrepresentation failure modes of (b) and (c).

3) E.3 Open-data policy: Locked: companion artifact `sq1_benchmark_floors_v2.md` published as CC-BY-4.0 open data alongside the paper. The paper’s negative claim is auditable only if the floor-sourcing is public. The companion artifact carries the full provenance table + URLs + access dates + primary sources.

4) E.4 Reproducibility statement: Locked (to be included in the paper at P6 if submission is to a venue that requires it): no from-laptop apparatus was run for this paper. All claims rest on external citations + published benchmark evaluations. Any reader can reproduce the gate result by (i) pulling the cited primary sources for each floor and candidate score, (ii) applying the Reading B band widths (Appendix B.2), (iii) running the aggregation rule (Appendix B.4). The companion artifact `sq1_benchmark_floors_v2.md` contains machine-readable sources

for mechanical re-execution.

5) E.5 Post-publication protocol: On any material change (≥ 5 pp) to a cited floor (Opus-4.7 or best ≤ 100 B) during the post-publication period:

1. The companion open-data artifact receives a dated errata entry with the new value preserved under §4 archive-never-delete.
2. The paper itself is not retracted. An errata pointer from the paper arxiv or lab-archive version to the updated companion is added.
3. If aggregate floor shift across all primary floors exceeds 5pp, a new program-open memo is drafted (per `question.md` kill criterion 6 as added by the amendment).

6) E.6 Escalation-trigger note: If at P6 the Director + PI decide to pursue a main-conference submission (NeurIPS, ICLR, ICML), the decision chain is:

1. Director opens a governance memo proposing lab-wide authorship convention (beyond the scope of this paper).
2. PI co-signs the convention after socialization period.
3. If the new convention retains option (a) for main-conference submissions, the paper is submittable under the current header; if the convention changes (e.g., to option (c) with a recruited human author), the paper must be re-emitted under the new header with the recruited human’s assent.
4. If no recruited human author is available or the convention is not resolved, the paper remains at the primary-workshop + arXiv + internal-archive tier. This is a complete-and-correct publication outcome; main-conference submission is not required for Program 1’s close.

Paper-writer escalates to Director if any C1-scope question emerges during the $v2 \rightarrow P6$ handoff that this appendix does not resolve.

O. References

Locked program artifacts (cited throughout):

- `programs/program_1_opus47_on_-18gb/question.md` (Phase 1 locked operational question).
- `programs/program_1_opus47_on_-18gb/unanimous_lock_li.md` (Phase 1 lock, D-109 remediation).
- `programs/program_1_opus47_on_-18gb/program_-open_memo.md` (program open; negative-result acceptance clause at :18-20).
- `programs/program_1_opus47_on_-18gb/prior_-work_1e.md` (compression-frontier and MoE-scale prior-work survey).
- `programs/program_1_opus47_on_-18gb/subquestions_1g.md` (SQ1 falsifiability clause; SQ1–SQ5 DAG).

- programs/program_1_opus47_on_18gb/scoping_1h.md (phase-by-phase plan; Alt-D branch).
- programs/program_1_opus47_on_18gb/tractability_1d.md (Phase 1 tractability assessment).
- programs/program_1_opus47_on_18gb/sq1_benchmark_floors.md (v1; archive-never-delete).
- programs/program_1_opus47_on_18gb/sq1_benchmark_floors_v2.md (v2; companion open-data artifact).
- programs/program_1_opus47_on_18gb/sq1_amendment_proposal.md (D-103/D-109).
- programs/program_1_opus47_on_18gb/sq1_amendment_addendum.md (D-104/D-109; red-team integration).
- programs/program_1_opus47_on_18gb/sq1_amendment_gate_criterion_supplement.md (D-105/D-109; Reading A/B/C).
- programs/program_1_opus47_on_18gb/p1_4_proxy_acknowledgment.md (three-band policy spec).
- programs/program_1_opus47_on_18gb/alt_d_envelope_paper_draft_v1.md (archived v1 draft, preserved per Values §4).
- programs/program_1_opus47_on_18gb/alt_d_envelope_paper_p5_red_team_report.md (P5 red-team REVISE verdict; basis for this v2).
- data/engineering/red_team_amendment_review_2026-04-17.md (adversarial review; Attacks 1–6).
- data/agents/pi/episodic/2026-04-17_program_1_phase_1_and_2.md (PI D-109 audit trail).
- data/values.md (§1, §4, §8, §9 cited).
- data/decisions_recent.md (D-103, D-104, D-105, D-109, D-110, D-111).

External technical references for the §5 compression-frontier survey (citations expanded beyond the outline per F2 requirement):

1. Frantar, E. et al. GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. arXiv:2210.17323, 2022. (inference-time INT4/INT3 PTQ on LLMs; cited for FP16 → INT4 MMLU-loss curve.)
2. Lin, J. et al. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. arXiv:2306.00978, 2023. (outlier-handling INT4 scheme; cited for 0.5–1pp FP16 → INT4 MMLU drop at 70B scale.)
3. Dettmers, T. et al. QLoRA: Efficient Finetuning of Quantized LLMs. arXiv:2305.14314, 2023. (NF4 quantization; cited for 4-bit-normal-float storage ± 0.5 pp loss.)
4. Xiao, G. et al. SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. arXiv:2211.10438, 2022. (activation-outlier smoothing; INT8 loss ≤ 0.5 pp.)

5. Ma, S. et al. The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits. arXiv:2402.17764, 2024. (BitNet b1.58 ternary-weight training-from-scratch; cited in §5.1 sub-4-bit discussion and §5.4 training-time compression.)
6. Shao, W. et al. OmniQuant: Omnidirectionally Calibrated Quantization for Large Language Models. arXiv:2308.13137, 2023. (2-bit PTQ with learnable calibration; 4–10pp MMLU drop at ≤ 70 B.)
7. Frantar, E. et al. SparseGPT: Massive Language Models Can Be Accurately Pruned in One-Shot. arXiv:2301.00774, 2023. (post-training unstructured pruning; combined with quantization costs 1–2pp MMLU.)
8. Sun, M. et al. A Simple and Effective Pruning Approach for Large Language Models (Wanda). arXiv:2306.11695, 2023. (N:M structured sparsity on LLMs; cited in §5.1 sparsity discussion.)
9. Fu, Y. et al. EAQuant: Expert-Aware Post-Training Quantization for MoE. arXiv:2506.13329, 2025. (MoE-specific INT4 PTQ; cited in §5.1 MoE-specific quantization.)
10. Staniszewski, K., Lancucki, A. KVTC: KV Cache Transform Coding. arXiv:2511.01815, ICLR 2026. (KV cache compression; cited in §5.3 context compression.)
11. Jiang, A. et al. Mixtral of Experts. arXiv:2401.04088, 2024. (Mixtral 8x7B; cited in §5.2 MoE at inference.)
12. DeepSeek-AI. DeepSeek-V3 Technical Report. arXiv:2412.19437, 2024. (DeepSeek V3 671B MoE; cited in §4.3 candidates + §5.2 MoE inference + footnotes ⁴ and ⁵ for DeepSeek V3.2 GPQA Diamond primary source.)
13. DeepSeek-AI. DeepSeekMoE: Towards Ultimate Expert Specialization. arXiv:2401.06066, 2024. (MoE routing architecture; cited in §5.2 MoE.)
14. Ludziejewski, K. et al. Joint MoE Scaling Laws: MoE Can Be Memory Efficient. arXiv:2502.05172, 2025. (MoE scaling laws; cited in §5.2 steelman for active-params.)
15. OpenAI. GPT-OSS Model Card. arXiv:2508.10925, 2025. (GPT-OSS 120B + 20B; cited in §5.2 MoE)

⁴DeepSeek V3.2 GPQA Diamond score pinned to primary source per C5 (red-team P5 report): DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024) reports 79.9 on GPQA Diamond for the V3 family under maj@k evaluation; the V3.2 variant inherits this evaluation per the DeepSeek-AI HuggingFace model card deepseek-ai/DeepSeek-V3.2 (accessed 2026-04-17). Where the V3 → V3.2 GPQA Diamond value was not separately re-reported, the V3 published figure is used as the active-params candidate headline, which is conservative-against-the-paper’s-conclusion (i.e., if V3.2 were worse, the gap would widen; if better, the gap would narrow but still hard-FAIL).

⁵DeepSeek V3.2 GPQA Diamond 79.9 primary source: DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024); HuggingFace model card deepseek-ai/DeepSeek-V3.2 (accessed 2026-04-17). See footnote at §4.3 for full discussion including the V3 → V3.2 inheritance argument.

- inference precedents.)
16. Microsoft Research. Sigma-MoE-Tiny Technical Report (20B/0.5B active). arXiv:2512.16248, 2025. (small-active MoE; cited in §5.2 active-params discussion.)
 17. Tencent. Hunyuan-Large (Hunyuan-MoE-A52B). arXiv:2411.02265, 2024. (Hunyuan MoE; cited as MoE candidate in §5.2.)
 18. Qwen Team. Qwen3 Technical Report. arXiv:2505.09388, 2025. (Qwen3 dense + MoE family; cited in §4.3 + §5.2.)
 19. Qwen Team. Qwen2.5 Technical Report. arXiv:2412.15115, 2024. (Qwen2.5-72B etc.; cited for §2.3 + §4.3 candidate scores.)
 20. Meta AI. Llama 3.3 70B Model Card (HuggingFace llama-3.3-70b-instruct). Meta AI, 2024. (Llama 3.3 70B benchmark numbers; cited for §4.3.)
 21. Benazir, M., Lin, J. Profiling LLM Inference on Apple Silicon. arXiv:2508.08531, 2025. (M3 Pro SSD bandwidth + inference latency; cited in §5.2 + Appendix D for storage-bound argument.)
 22. Kashani, V. metalQwen3: Metal GPU Transformer Implementation. QNeura.ai, 2025. (Metal inference on Qwen3; cited for §5.2 Apple Silicon realizability.)
 23. Multiple Authors. KPOP: Training on Apple Silicon. ICML 2025 workshop. (Apple Silicon training feasibility; cited for §5.4 from-laptop training scope.)
 24. Muennighoff, N. et al. Scaling Data-Constrained Language Models. arXiv:2305.16264, 2023. (data-constrained scaling; cited in §5.4 training-from-scratch envelope.)
 25. Rein, D. et al. GPQA: A Graduate-Level Google-Proof Q&A Benchmark. arXiv:2311.12022, 2023. (GPQA / Diamond benchmark paper; cited in §4.2.)
 26. Wang, L. et al. Auxiliary-Loss-Free Load Balancing for MoE (DeepSeek). arXiv:2408.15664, 2024. (MoE load balancing; cited in §5.2 MoE routing.)
 27. (Removed in v2 per C3 — former “DeltaMoE” citation lacked arXiv ID + authors + venue metadata in data/bibliography.md and could not be verified externally. Discussion of expert-offload / delta-compression schemes in §5.2 + Appendix D.2 retains MoE-Offload and ExpertFlow references; see ref 30.)
 28. Various. FlashMGLU: Masked Gated Linear Unit. arXiv:2506.23225, NeurIPS 2025. (fast MGLU kernels; cited in §5.2 MoE kernel efficiency.)
 29. Lu, W. et al. Not All Experts are Equal: Efficient Expert Pruning. 2024. (MoE expert pruning; cited in §5.1/§5.2 expert compression survey.)
 30. He, Y. et al. ExpertFlow: Efficient MoE Inference. 2024. (expert offload inference; cited in §5.2 offload latency + Appendix D.2 active-params steelman.)
 31. Panda, S. et al. Dense Backpropagation Improves Training for Sparse MoE (Default MoE). arXiv:2504.12463, 2025. (MoE training efficiency; cited in §5.4 from-laptop training.)
 32. Thérien, B. et al. Continual Pre-training of MoEs: How robust is your router? arXiv:2503.05029, 2025. (MoE router robustness under continual training; §5.4 distillation discussion.)
 33. Yan, Z., Bai, X., Yao, Z., Liu, D., Liu, T., et al. Scalable Training of Mixture-of-Experts Models with Megatron Core (NVIDIA). arXiv:2603.07685, 2026. (MoE training infrastructure; §5.4 training-time compression reference; cited inline as “Yan et al., 2026” for chronological disambiguation per C2.)
 34. Wang, X., Hayou, S., Nalisnick, E. The Myth of Expert Specialization in MoEs: Why Routing Reflects Geometry, Not Necessarily Domain Expertise. arXiv:2604.09780, April 2026. (MoE routing theory; §5.2 failure-mode reference on expert homogenization and specialization; cited inline as “Wang et al., 2026” for chronological disambiguation per C2.)
- ARC
<https://arcprize.org/leaderboard>

HuggingFace
https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard

llm-stats.com: <https://llm-stats.com/models/claude-opus-4-7>

Vellum model cards: <https://www.vellum.ai/models/claude-opus-4-7>

HuggingFace model card DeepSeek V3.2: <https://huggingface.co/deepseek-ai/DeepSeek-V3.2>

ICBINB workshop (methodology-literature anchor, §2.5): I Can’t Believe It’s Not Better workshop at NeurIPS, proceedings 2020–present. (Negative-result methodology canon; cited in §2.5 as methodology anchor.)

ML Reproducibility Challenge (methodology-literature anchor, §2.5): MLRC proceedings 2018–present. (Reproducibility-crisis methodology; cited in §2.5.)

Prize
(ARC-AGI-v2 public and semi-private scores; accessed 2026-04-17).

OpenLLM
(Llama 3.3 70B, Qwen2.5-72B, QwQ-32B-preview scores cross-referenced; accessed 2026-04-17).

Leaderboard:
(Opus-4.7 aggregator row; accessed 2026-04-17).

(Opus-4.7 aggregated model card; accessed 2026-04-17).

DeepSeek V3.2 model card; accessed 2026-04-17; cited in footnotes ⁶ and ⁷ per C5).
-
- ⁶DeepSeek V3.2 GPQA Diamond score pinned to primary source per C5 (red-team P5 report): DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024) reports 79.9 on GPQA Diamond for the V3 family under maj@k evaluation; the V3.2 variant inherits this evaluation per the DeepSeek-AI HuggingFace model card deepseek-ai/DeepSeek-V3.2 (accessed 2026-04-17). Where the V3 → V3.2 GPQA Diamond value was not separately re-reported, the V3 published figure is used as the active-params candidate headline, which is conservative-against-the-paper’s-conclusion (i.e., if V3.2 were worse, the gap would widen; if better, the gap would narrow but still hard-FAIL).

⁷DeepSeek V3.2 GPQA Diamond 79.9 primary source: DeepSeek-AI, DeepSeek-V3 Technical Report (arXiv:2412.19437, 2024); HuggingFace model card deepseek-ai/DeepSeek-V3.2 (accessed 2026-04-17). See footnote at §4.3 for full discussion including the V3 → V3.2 inheritance argument.

P. Acknowledgments

This paper is the collective output of the AGI Lab autonomous agent organization. The Director role (session 2026-04-17) dispatched the phase sequence; the chief_scientist role drafted the section outline (alt_d_paper_outline.md); the paper_writer role (this draft, v1 + v2) expanded to full prose and applied the P5 red-team revision pass (v1 → v2); the measurement_theorist role produced the floor pinning and gate re-run (sql_benchmark_floors_v2.md); the red_team role produced two adversarial reviews (the amendment-package review red_team_amendment_review_2026-04-17.md and the P5 paper-draft review alt_d_envelope_paper_p5_red_team_report.md); the evaluator role produced the closure verdict; the PI role provided D-109 co-sign under Reading B. The lab's values file (data/values.md) §9 framing of aspirational missions is the intellectual foundation for the paper's interpretation of the negative result. The procedural and semantic files under data/agents/ provide the operational framework under which each role acted. No external humans contributed content to this paper; all agent roles are instances of Anthropic's Claude Opus 4.7 model operating under role-specific semantic and procedural prompts per the lab's D-109 remediation standard. The human submitter of record for external-facing venues (Harshith Kantamneni, hkantamneni2@wisc.edu, UW-Madison) serves an administrative account-holder role per Appendix E.1 and does not claim authorship.

Q. Sign-off and status

paper_writer signature: ✓ (2026-04-18, P5 draft v2). P5 reviewers: red_team (adversarial review of draft v1 — REVISE verdict; see alt_d_envelope_paper_p5_red_team_report.md); evaluator (closure-verdict audit — pending on v2 verification of C2 arXiv URL resolution, C5 DeepSeek V3.2 primary source). P6 signers (pending): Director (synthesis + dispatch close); PI (co-sign + §9 framing confirmation).

Status: DRAFT v2 complete. All 6 required P5 red-team edits (C1–C6) applied; both optional edits (C7–C8) applied. Full change log in front-matter. v1 archived in place at alt_d_envelope_paper_draft_v1.md per Values §4 archive-never-delete. All 8 body sections and all 5 appendices (A–E) are populated; Appendix A row 4 has the inlined primary-source URL per C4; Appendix E.1/E.2 expanded with venue-fallback table + authorship convention lock per C1; §2.5 Related Work added per C7; §6.5 closing sentence tightened per C8; §5.5 compounding-math carries no-empirical-run parenthetical per C6; refs 33 + 34 have author-year parentheticals at body citations per C2; ref 27 removed and Appendix D.2 rewritten per C3; DeepSeek V3.2 primary source footnoted at §4.3 + Appendix D.4 per C5.

Meta-pre-commitment compliance check for v2: the revision does not propose any SQ1 floor relaxation, does not propose any eligibility-criterion amendment, does not smuggle a Program-1-lite demo into §5, does not modify the meta-pre-commitment itself, does not reframe the gate-FAIL closure. The closure claim in §8.1 is the negative-result finding; no pivot framing is offered. If P5 evaluator verification or P6 signer review requires relaxing floors to make a story work, paper_writer escalates to Director rather than softening — the meta-pre-commitment binds this paper per sql_amendment_addendum.md §4 item 9. Meta-pre-commitment audit: HELD (same verdict as v1; v2 introduces no change that risks violation).

End of draft v2.

#	Benchmark	Opus-4.7 floor	Source type	Primary source	Corroborating sources	Protocol	Confidence	Proxy error-bar ($\pm$ pp)
1	MMLU (5-shot)	91.0%	Proxy-inherited from sibling models (Claude Opus 4.5/4.6 disclosure ranges)	Anthropic model-card materials for Claude Opus 4.5 + 4.6 (MMLU disclosed as range 88.7–91.4); extrapolation to 4.7 under monotonic sibling-model assumption	HuggingFace OpenLLM leaderboard (historical entries); HELM MMLU report on Claude family	5-shot zero-temperature single-sample	Medium (proxy; no direct 4.7 MMLU publication)	± 1.5 pp (proxy + saturation)
2	HumanEval (pass@1)	96.8%	Third-party reported (non-vendor publication)	Startup Fortune Claude 4.7 launch coverage reporting 96.8 pass@1 HumanEval; repeated in llm-stats and Vellum aggregators	Vellum model-card row; llm-stats.com Opus-4.7 entry	pass@1, single-sample reported	Medium-High (third-party; single-primary-source)	± 1.0 pp (near-ceiling noise)
3	GSM8K (maj@1 or 5-shot CoT)	96.0%	Proxy-inherited from saturation argument + 4.5/4.6 disclosure	Anthropic materials for 4.5/4.6 showing GSM8K near-saturation at ~95–96; extrapolation to 4.7 under saturation-floor assumption	Vellum aggregation; open-weight Qwen2.5-72B and Llama 3.3 70B published ≥ 95 GSM8K corroborating the near-saturated frontier	5-shot CoT with majority vote or single-sample	Medium (proxy + saturation near ceiling)	± 2.0 pp (saturation + CoT variance)
4	GPQA Diamond	94.2%	Direct Anthropic publication + multi-source corroboration	Anthropic launch/system-card materials for Claude Opus 4.7 (headline figure 94.2% GPQA Diamond). Primary source URL: https://www.anthropic.com/news/claude-opus-4-7-model-page (accessed 2026-04-17). Archival snapshot: Vellum model card at https://www.vellum.ai/models/claude-opus-4-7 (accessed 2026-04-17, reports 94.2% GPQA Diamond). llm-stats aggregator: https://llm-stats.com/models/claude-opus-4-7 (accessed 2026-04-17, reports 94.2% GPQA Diamond).	Vellum model-card aggregation (94.2% Opus-4.7 Diamond row); llm-stats.com Opus-4.7 entry (94.2% Diamond); VentureBeat article: https://venturebeat.com/2025/04/17/anthropic-launches-4-7-model/ (accessed 2026-04-17, reports 94.2% GPQA Diamond). Decrypt launch coverage (all reporting 94.2 vs Opus-4.6’s 91.3)	maj@k extended thinking per Anthropic 4-family methodology footnote	High	± 3.0 pp (Reading B band — TruthfulQA analogy + binomial SE at N=198)

Tier	Venue	Authorship policy	Lab's operational path	Fallback if rejected
Primary (honest-negative-result workshops)	ICBINB ("I Can't Believe It's Not Better," NeurIPS workshop)	Permissive; accepts collective authorship with contributor statement	Submit with authorship line "AGI Lab (autonomous agent collective), Program 1, Alt-D branch" + contributor-statement listing agent roles; human submitter-of-record: Harshith Kantamneni (hkantamneni2@wisc.edu, University of Wisconsin-Madison)	If ICBINB chairs reject collective-authorship format, re-submit with human submitter listed as author-of-record and agent roles listed in contributor statement; do NOT re-submit with a silently-introduced "figurehead human author"
Primary	MLRC (ML Reproducibility Challenge)	Permissive; methodology + reproducibility papers with author-team flexibility	Same as ICBINB	Same fallback as ICBINB
Primary	"Don't Blame the Data" workshop	Permissive; aligned with negative-result methodology	Same as ICBINB	Same fallback as ICBINB
Robust fallback	arXiv preprint (cs.LG / cs.AI)	Requires named submitter; authorship format at author's discretion subject to arXiv guidelines	Submit under authorship "AGI Lab (autonomous agent collective), Program 1, Alt-D branch"; submitter Harshith Kantamneni (hkantamneni2@wisc.edu) as account-holder-of-record. arXiv's submitter-of-record field is administrative; it does not imply authorship under the lab's convention.	If arXiv metadata rejects collective authorship in the Authors field, use "AGI Lab" as the org-level author-string and include the full agent-role list in the Abstract + first section footnote
Robust fallback	Internal lab archive (programs/program_1__opus47_on_18gb/ under the AGI repo)	Lab policy; D-109 recognizes agent authorship with role-by-role contribution	Archive the v2 paper + companion open-data artifact here indefinitely; this is the authoritative long-term lab-internal version	N/A — lab-internal archive is not subject to external authorship policy
Aspirational (requires P6 human PI co-sign)	NeurIPS main track	Requires human authors with institutional affiliations; does not currently accept fully-AI-authored submissions	Paper-writer scope ends here; any main-track submission requires (i) explicit P6 decision by Director and PI, (ii) recruitment of a human author-of-record willing to take ethics-statement responsibility, (iii) Director socialization of authorship convention across the lab before commitment	Fallback is to keep the paper at the primary-workshop + arXiv + internal-archive tier; do NOT attempt a main-conference submission without P6 human-author decision
Aspirational (requires P6 human PI co-sign)	ICLR main track	Same as NeurIPS main track	Same as NeurIPS main track	Same fallback
Aspirational (requires P6 human PI co-sign)	ICML main track	Same; does not accept fully-AI-authored submissions	Same as NeurIPS main track	Same fallback